TurboWin+ version 4.4 Noteworthy features and Tips

1 Email options

1.1] General

Two settings (address recipient and subject) are always required before you can send observations by TurboWin+ (Maintenance → Email settings).

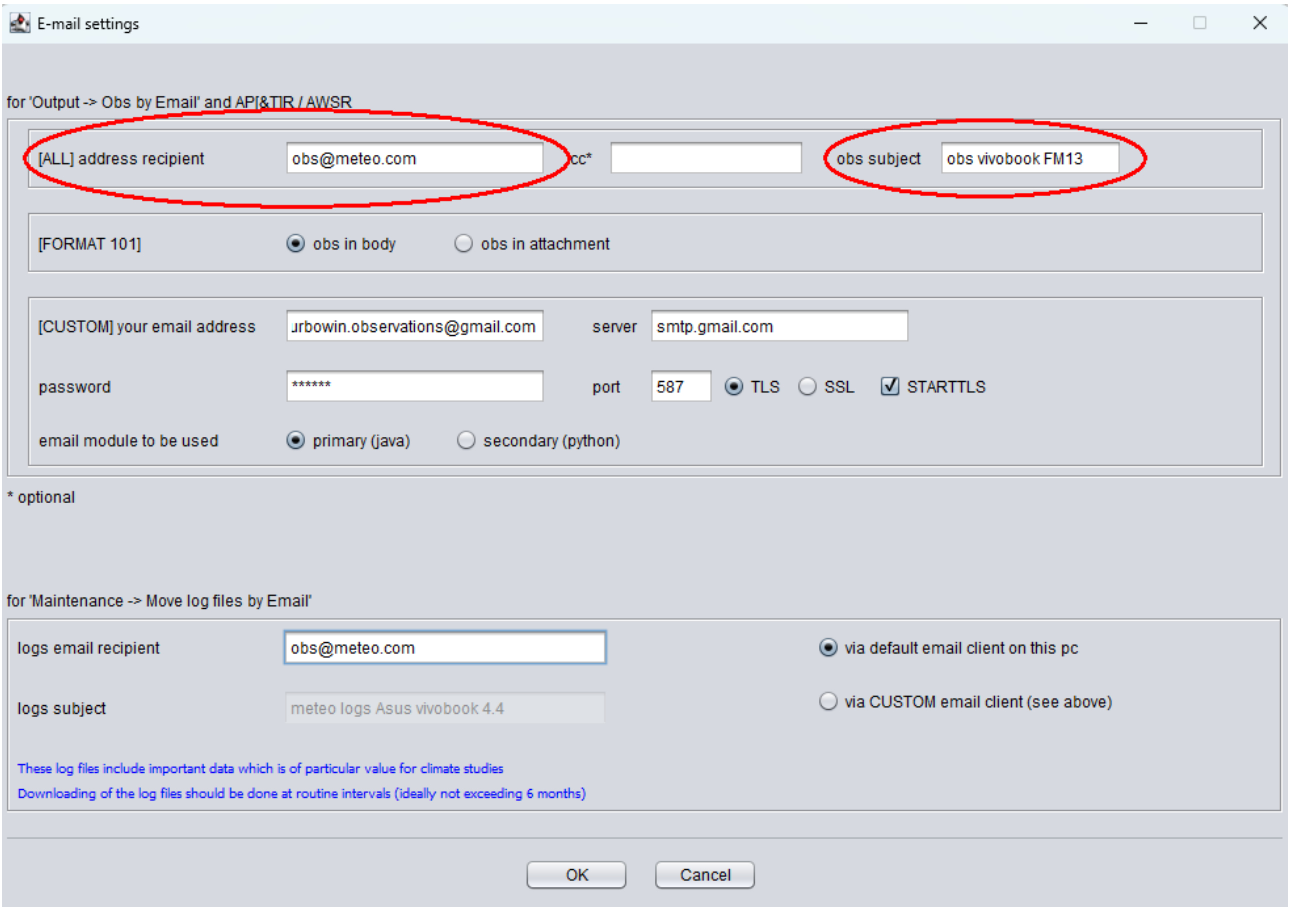

fig 1 Email settings example with required basic data surrounded in red

1.2] Setting up (SMTP) local host for TurboWin+ Custom email

Sending via local host [Output – Obs by Email (Custom)] could be an alternative for sending via the default email app. Ask the system administrator for the required settings.

1.3] Setting up Gmail for TurboWin+ Custom email

basically this are the required steps

1. If no account create a new account [https://support.google.com/mail/answer/56256?hl=en]
2. For this account turn on 2-step verification [https://support.google.com/accounts/answer/185839]
3. Generate an app password [e.g. https://www.lifewire.com/get-a-password-to-access-gmail-by-pop-imap-2-1171882] [see note 1 below and fig 3]
4. Insert your Gmail address into the TurboWin+ Email settings form [fig 4]
5. Insert the Gmail server name into the TurboWin+ Email settings form [fig 4]
6. Paste the (app) password into the (app) password field in the TurboWin+ Email settings form [see note 2 below and fig 4]
7. Insert the appropriate port number into the TurboWin+ Email settings form [see note 3 below and fig 4]
8. Select the appropriate security settings (TLS, SSL, STARTTLS) in the TurboWin+ Email settings form [see note 3 below and fig 4]
9. Select the email module to be used (primary or secondary) [see note 4 below and fig 4].

Note 1

The value of an application-specific password is that you can revoke and regenerate a password on a service-by-service basis instead of having to change the master password to your account. If you do need to create a new app-specific password for a program or service, revoke passwords previously set up but no longer used for the same application.

It's best practice to use an app-specific password only for a single service. You are free to generate as many app-specific passwords as you like. [source: https://www.lifewire.com/get-a-password-to-access-gmail-by-pop-imap-2-1171882]

Note 2

Most of the time, you'll only have to enter an (app) password once per app or device, so don't worry about memorizing it. You can also always generate a new one.

Note 3

TLS uses port 587 and SSL port 465. One or both ports could be initially closed. TLS is recommended. Ask the system administrator

Note 4

Why 2 email options?

advantage primary email module: The new implemented email module is written in Java and fully integrated in the TurboWin+ main program. No virus scanner and firewall issues

disadvantage primary email module: Compared to the secondary email option the transport layer security protocol (https://en.wikipedia.org/wiki/Transport_Layer_Security) must be set in the source code, which may cause problems with certain email providers in the future.

advantage secondary email module: No settings of the transport layer protocol set up in the source code needed.

disadvantage secondary email module: Still a distinctive program, new process will be spawn. This can e.g. lead to problems with the virus scanner/firewall

Since sending observations by email is still a much commonly used method, it was decided to offer both options as maximum flexibility for the users on board

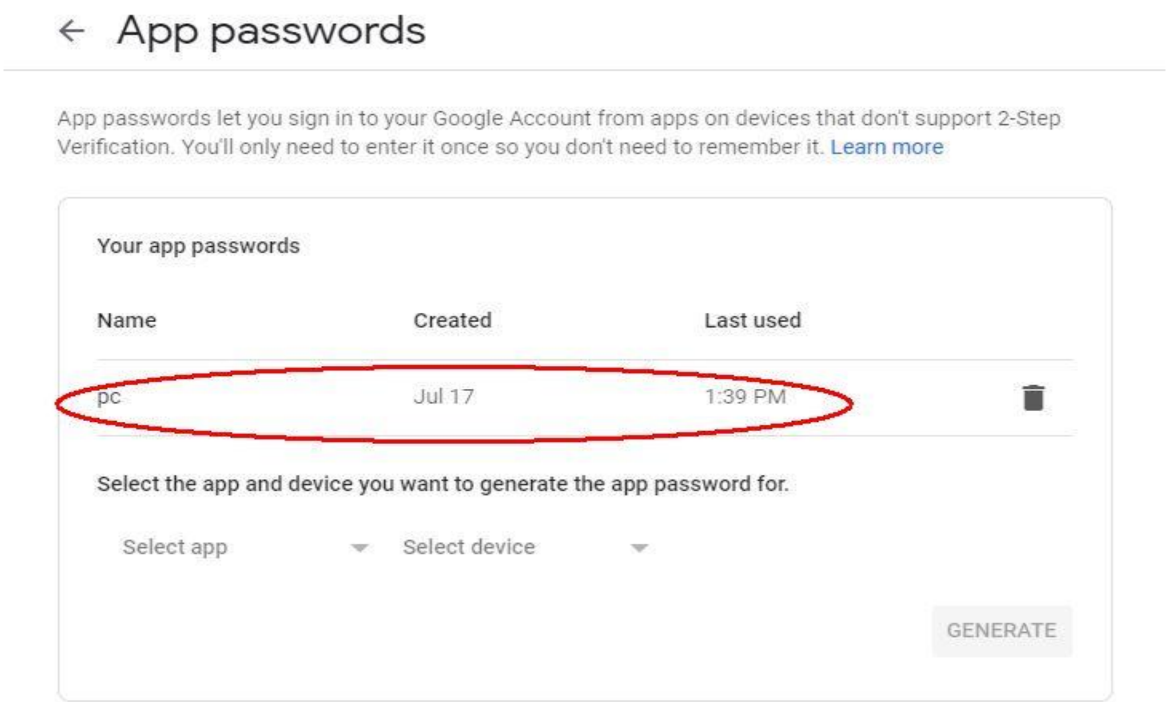

fig 3 View of app passwords on Google account web site for (test)email address turbowin.observations@gmail.com

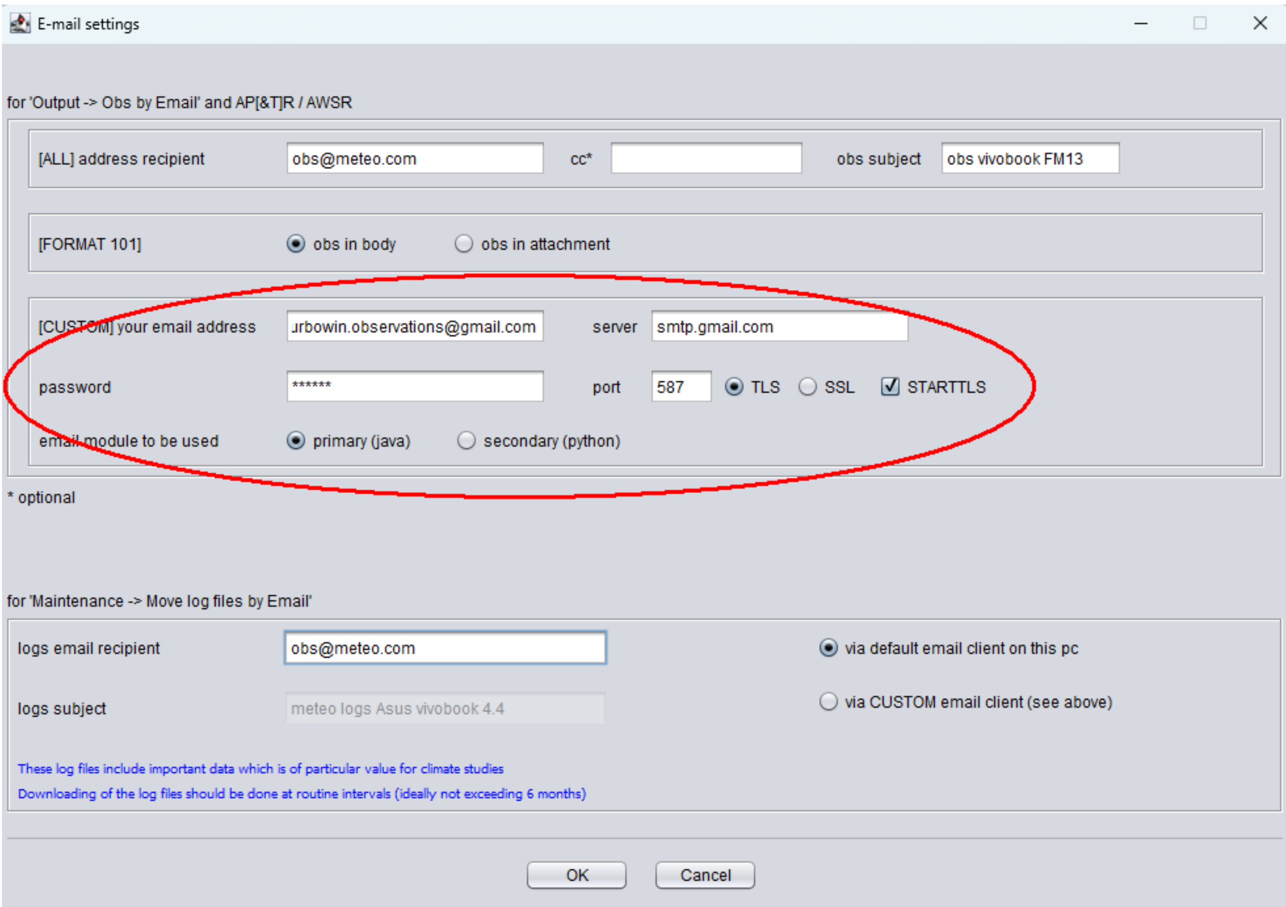

fig 4 example, setting up Gmail for TurboWin+

1.4] Setting up Yahoo Mail for TurboWin+ Custom Email

basically this are the required steps

1. If no account create a new account [https://login.yahoo.com/account/create?.lang=en-US&.intl=us&.src=yhelp]

2. For this account turn on 2-step verification [https://help.yahoo.com/kb/SLN5013.html]

3. Generate an app password [https://uk.help.yahoo.com/kb/account/generate-third-party-passwords-sln15241.html] [see note 1 below and fig 6 and fig 7]

4. Insert your Yahoo address into the TurboWin+ Email settings form [fig 8]

5. Insert the Yahoo server name into the TurboWin+ Email settings form [fig 8]

6. Paste the (app) password into the (app) password field in the TurboWin+ Email settings form [see note 2 below and fig 8]

7. Insert the appropriate port number into the TurboWin+ Email settings form [see note 3 below and fig 8]

8. Select the appropriate security settings (TLS, SSL, STARTTLS) in the TurboWin+ Email settings form [see note 3 below and fig 8]

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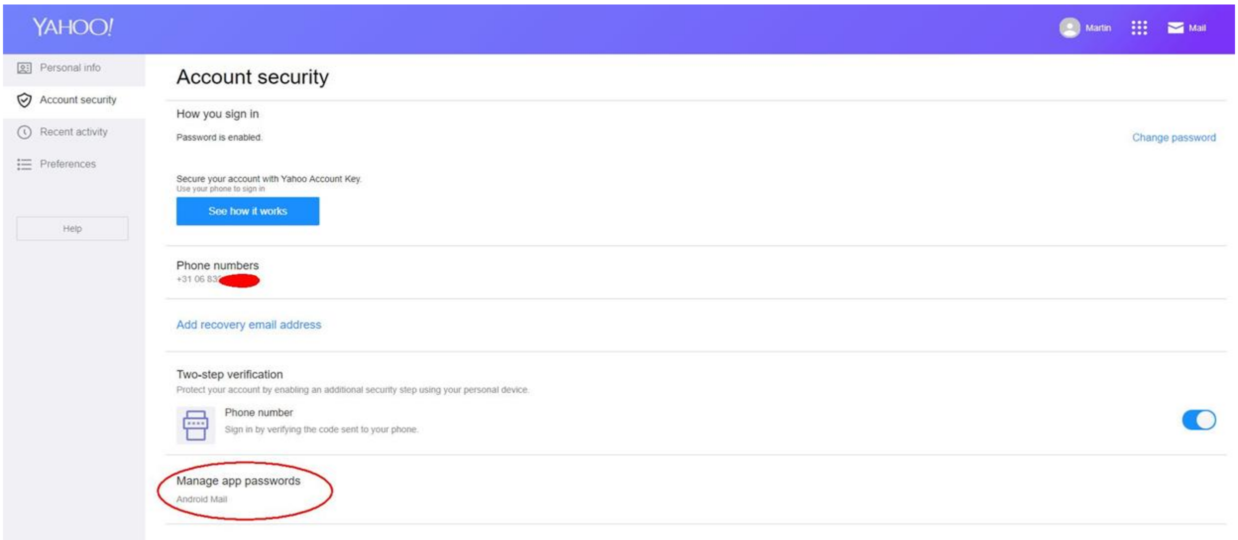


fig 6 setting up Yahoo email for TurboWin+ (Yahoo account web site)

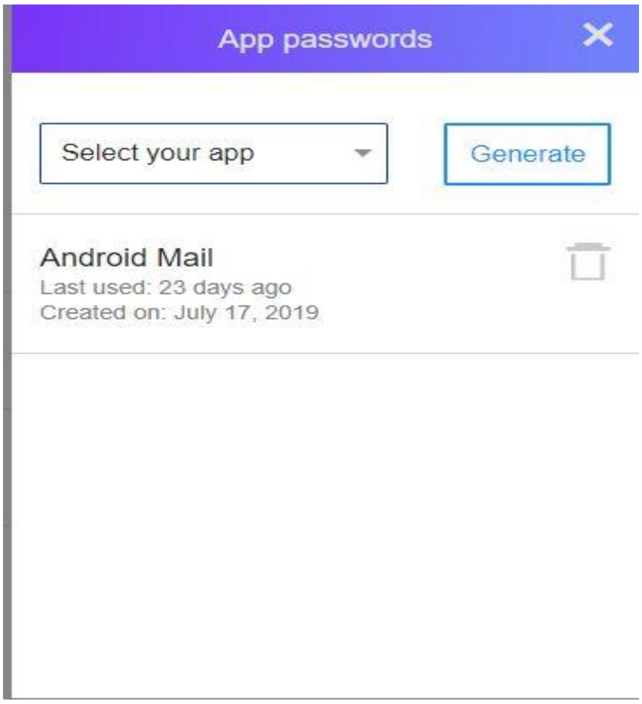


fig 7 View of app passwords on Yahoo account web site for (test)email address turbowin@yahoo.com

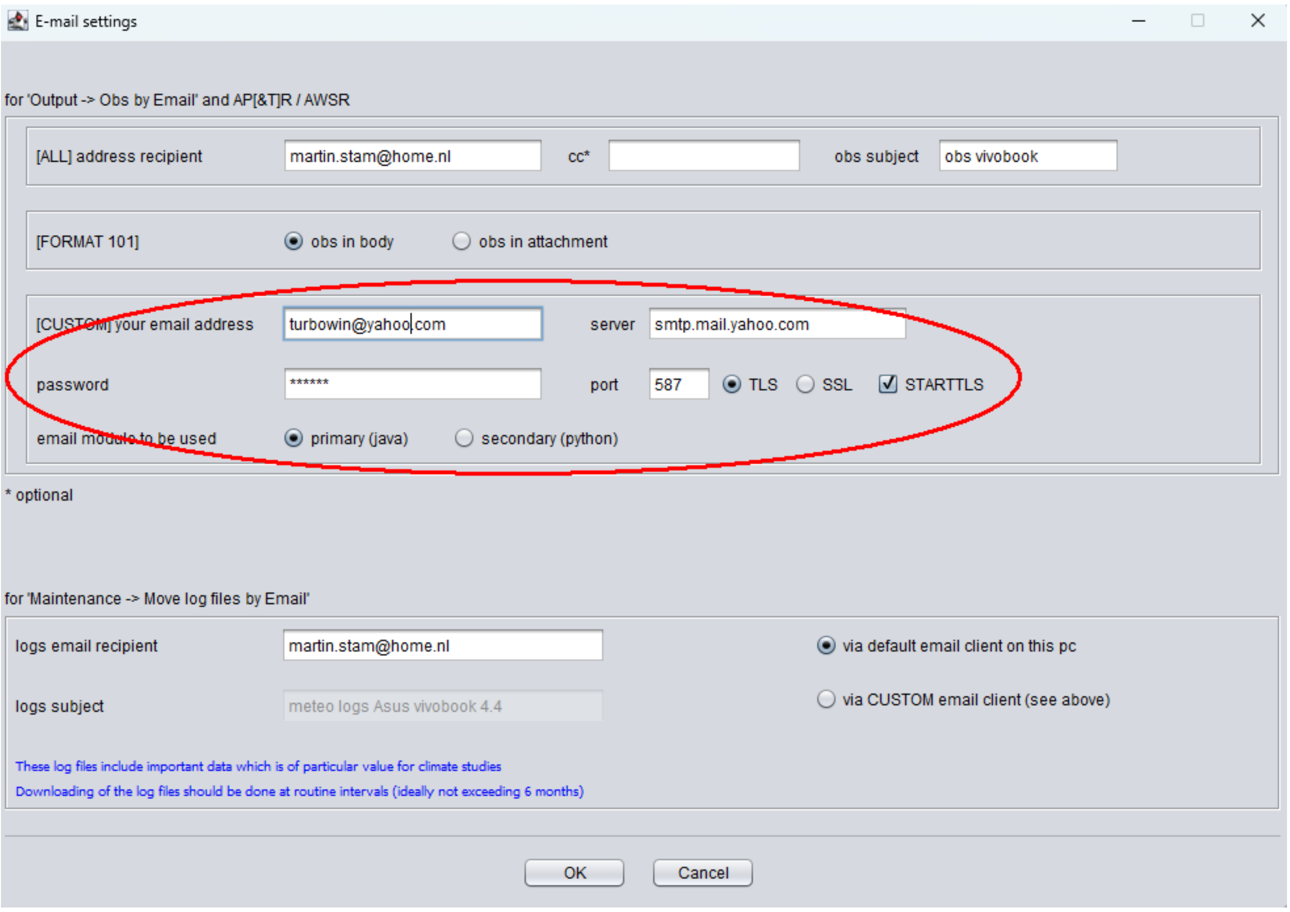


fig 8 example, setting up Yahoo Mail for TurboWin+

1.5 Sending and Logging

In manual mode: an Email output option (fig 10) will only be available (not grayed out) if the required settings for that specific option are made before in the Email settings form (fig 1).

note

In APR, APTR and AWSR mode all output options are not available (grayed out) by default. If you turn off APR (APTR or AWSR) the appropriate output options will be selectable again. In EUCAWS mode only the output option “Obs to AWS” is available for adding manual observation data

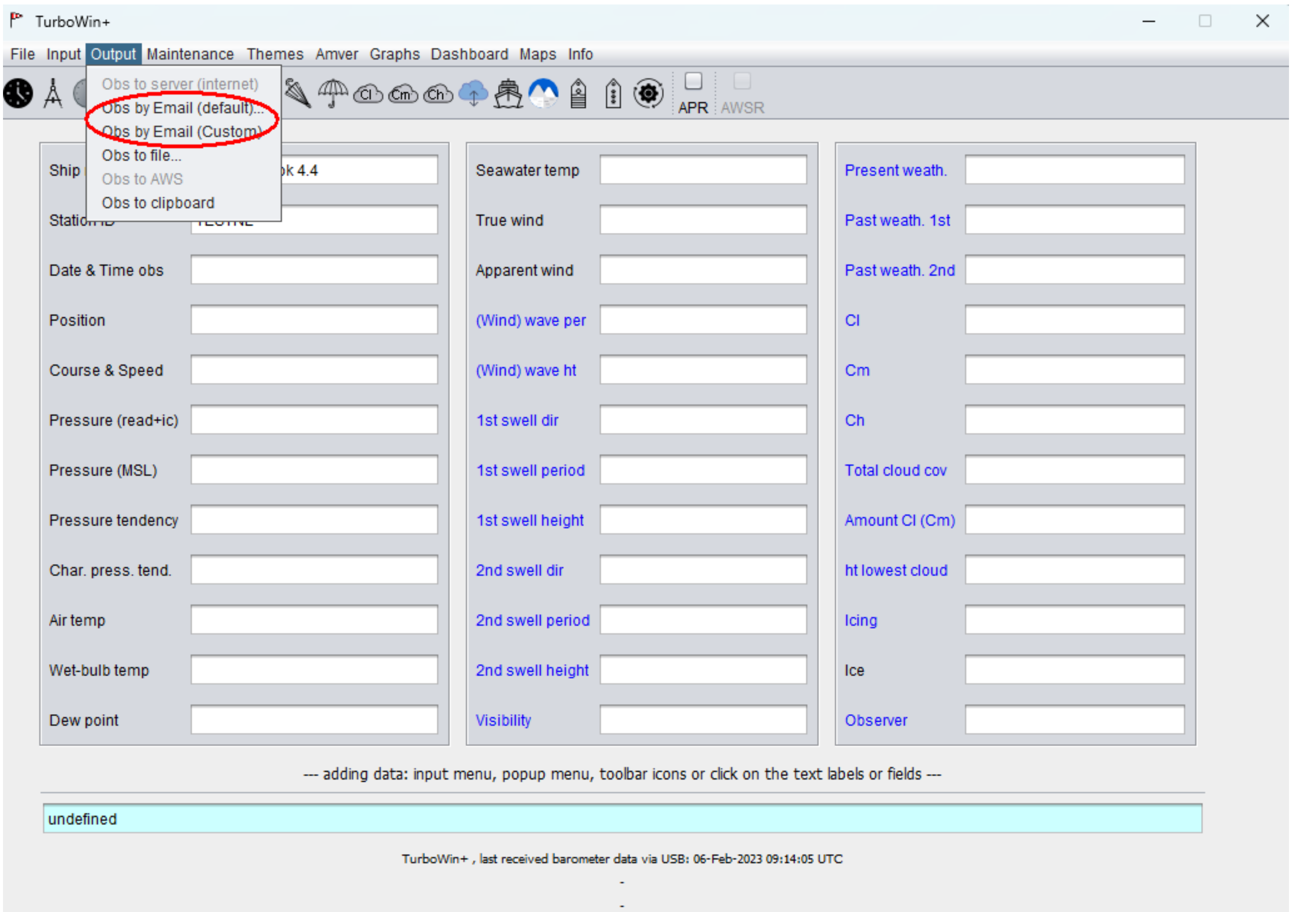


Fig 10 Email output options (surrounded in red); with the email output options available

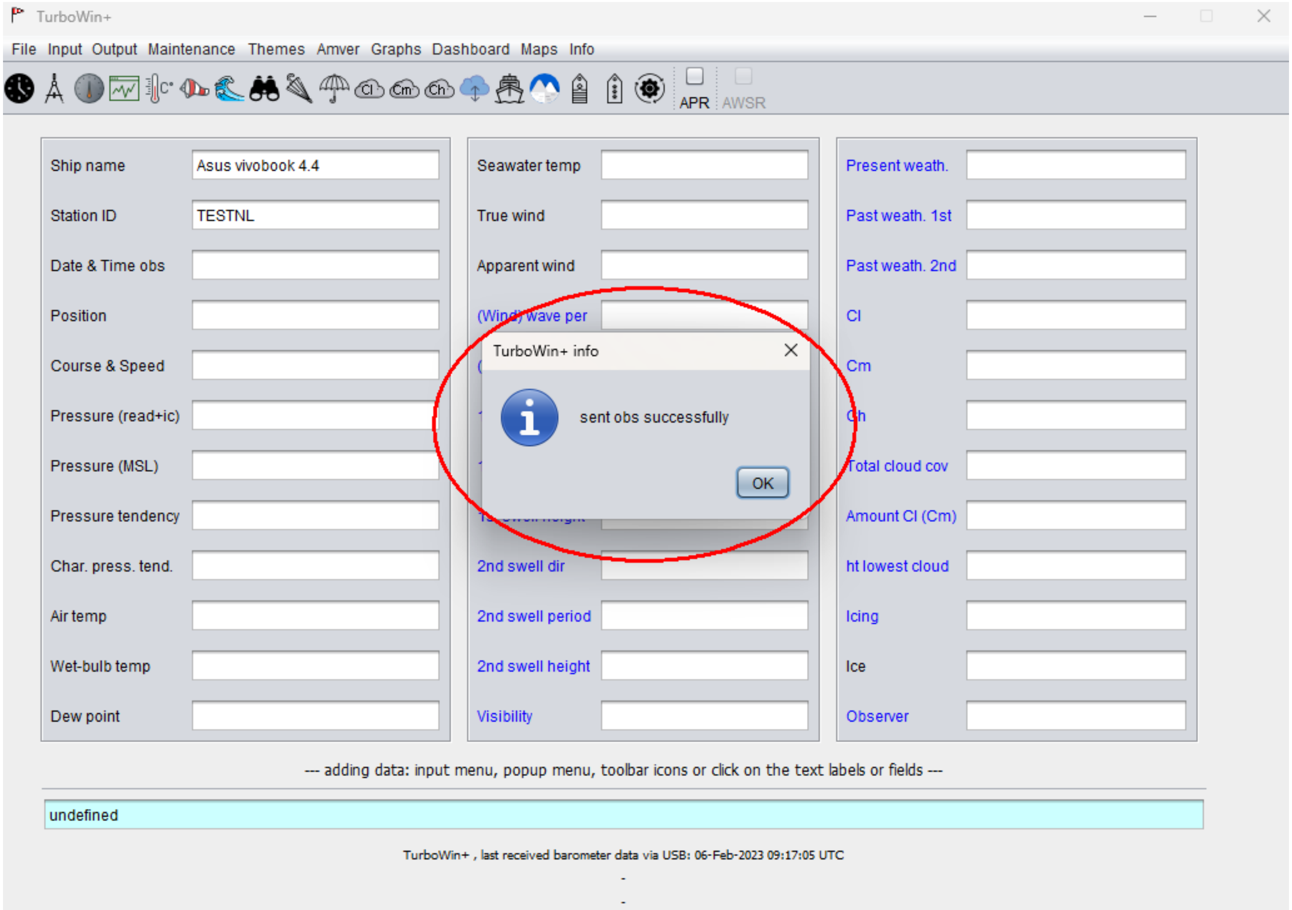


Fig 11 pop-up feedback after sending an observation by Email

```
06-Feb-2023 09:17:49 UTC [GPS] last position parsing ok; GPS position (dd-mm [N/S] ddd-mm [E/W]): 52-42 N 6-14 E
06-Feb-2023 09:17:58 UTC [EMAIL] trying to send obs (body= "BBXX TESTNL 0612/ 99527 10062 41/// ///// 1//// 2//// 4//// 5//// 7//// 8//// 22200 0//// 2////=") to ma
06-Feb-2023 09:17:59 UTC [EMAIL] obs email sent successfully
06-Feb-2023 09:17:59 UTC [IMMT] appended immt.log successfully
```

fig 12 Extended logging (TurboWin+: Info → System log)

2 Vaisala HMP155 sensor interface

For air temperature, wet-bulb temperature, dew point and relative humidity

Fig 14 Vaisala HMP155

On Windows 7 the required Vaisala HMP155 USB driver (called: “Vaisala USB device”) must be installed manually (from the accompanied Vaisala cd or from the Vaisala web site). It seems that on most Windows 10 pc’s the driver will be installed automatically. Maybe you have to restart the pc after inserting the HMP155 USB cable. On Linux no additional driver required.

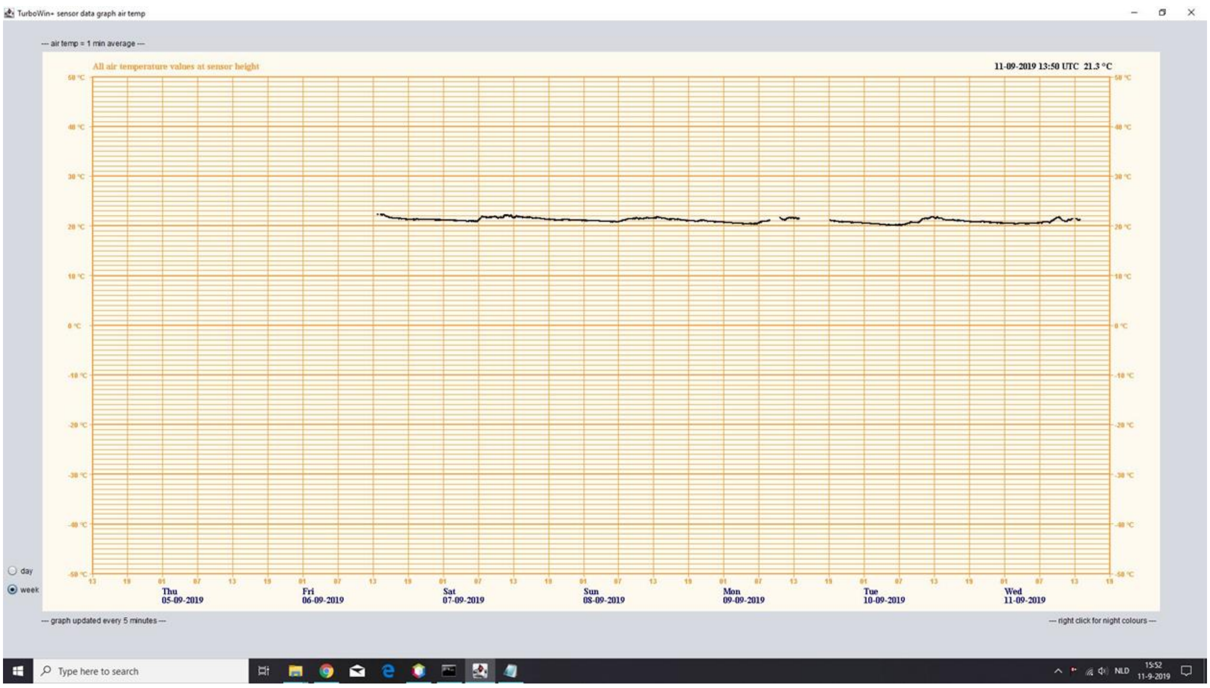


Fig 15 TurboWin+ air temperature graph of measured Vaisala HMP155 data

The HMP155 can be used as one of the required sensors in APTR (Automated Pressure & Temperature reports) mode and manual mode. Observation format #101 or format FM13 via server or email

3 EMOS (Mintaka) interface

-

fig 16 The main components of EMOS: Mintaka Star + StarX + GuardX; For GPS position, air pressure, pressure tendency and characteristic, air temperature, wet-bulb temperature, dew point and relative humidity

EMOS (Enhanced Manual Observation System) can be used in APTR (Automated Pressure & Temperature reports) mode and manual mode. Connection via WiFi or USB. Observation format #101 or format FM13 via server or email.

4 Setting TurboWin+ to APR (or APTR)

4.1 settings

- Select the observation format (Maintenance -> Obs format setting, Fig 17)
- Prepare the output send/upload options (Maintenance -> Email settings or Maintenance → Server settings)
- Select the connected sensors (Maintenance -> Serial/USB/LAN device settings, Fig 19)
- Set the APR options (Maintenance -> APR/APTR/AWSR settings, Fig 20)
- Optional: set the visual reminder pop-up screen (Fig 20b and Fig 20c)

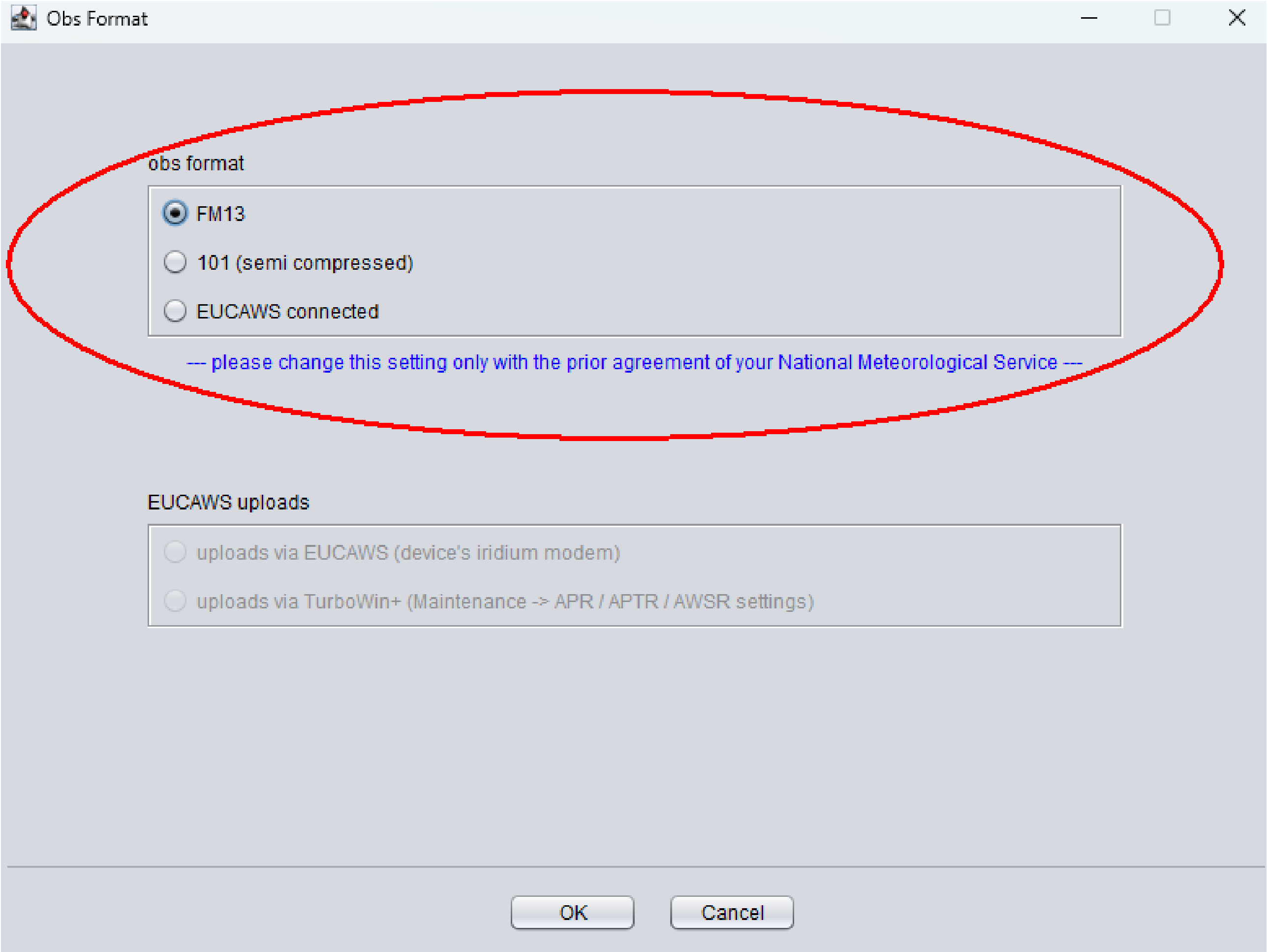


fig 17 selecting obs format

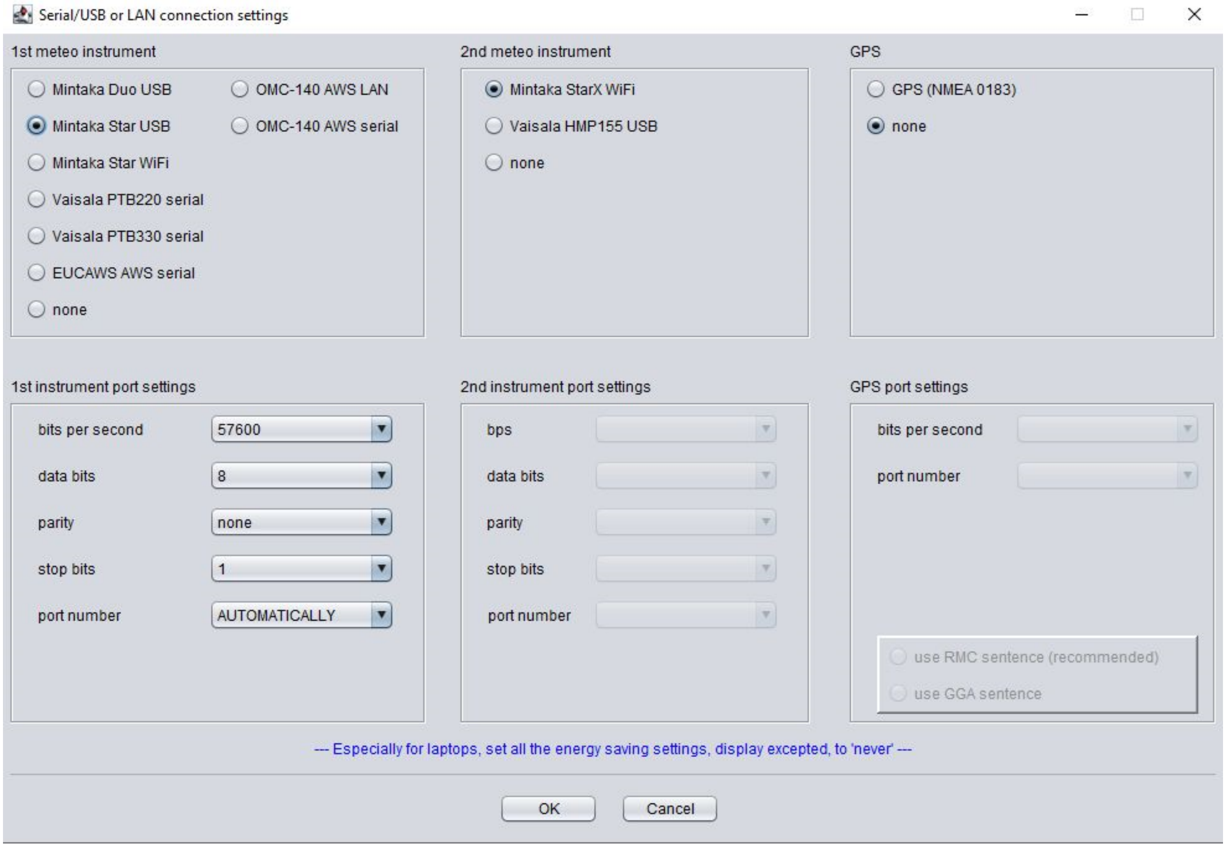


fig 19 selecting connected sensors

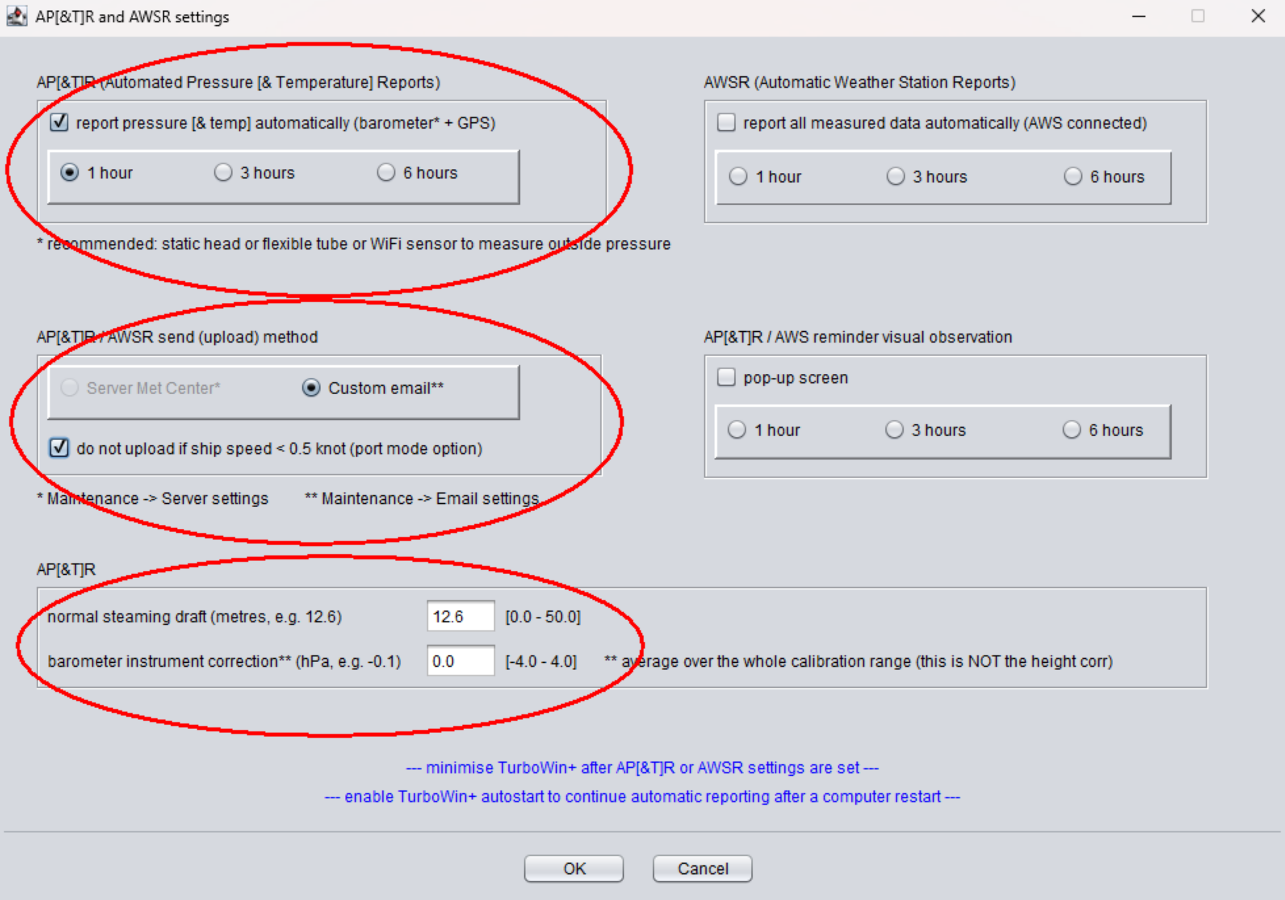


fig 20 selecting the specific APR (APTR) options

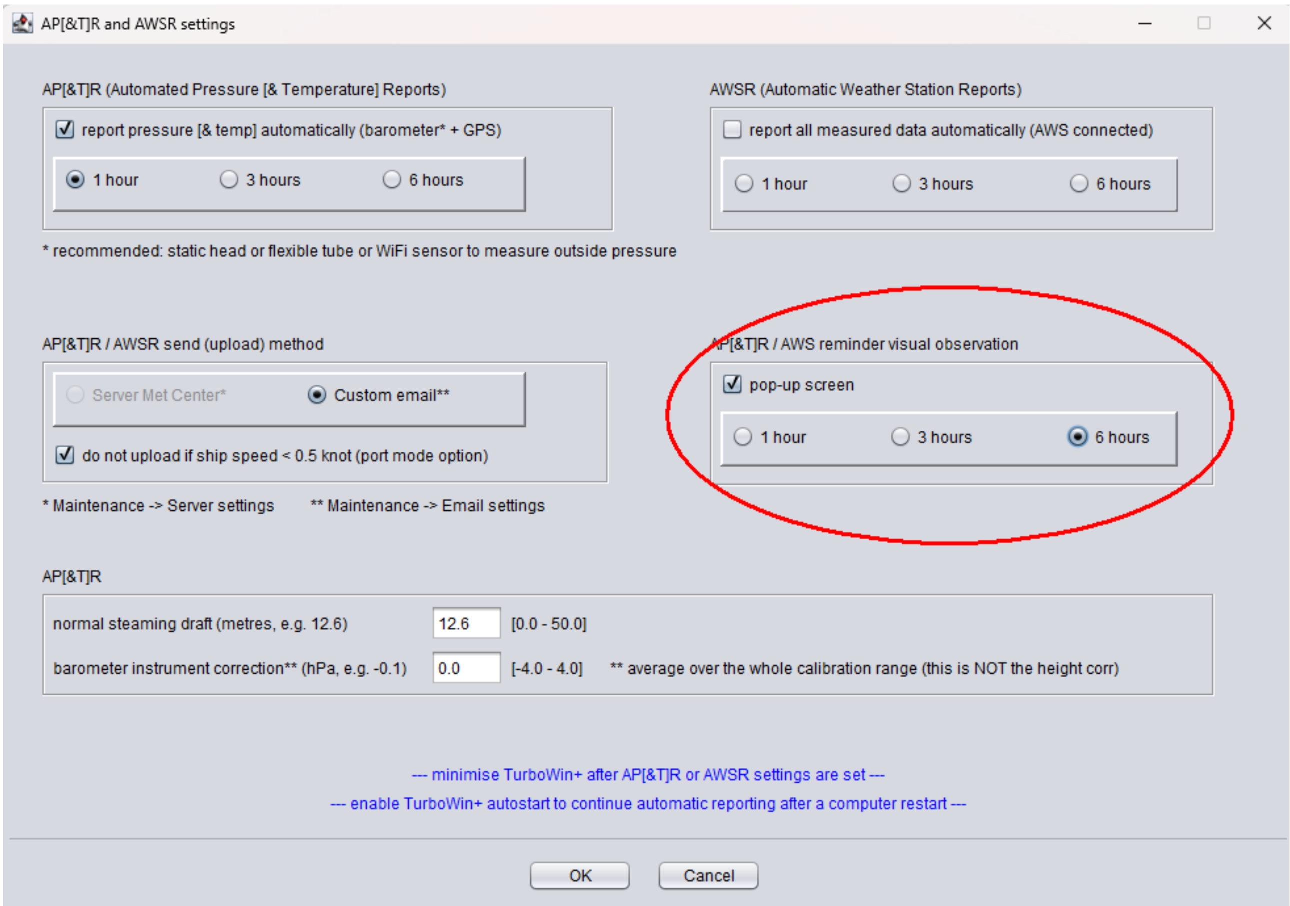


fig 20b selecting the extra visual reminder option

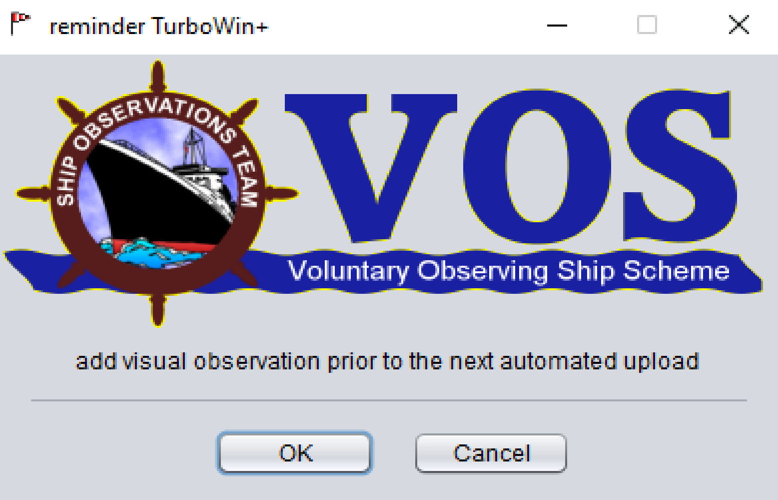


fig 20c visual reminder pop-up

4.2 Adding manual observed data when in APR/APTR mode

This is very easy. Open the appropriate meteorological parameter input screen, add the data and click OK (Fig 21). No further actions required. The inserted data will be automatically added to the observation with the next scheduled upload.

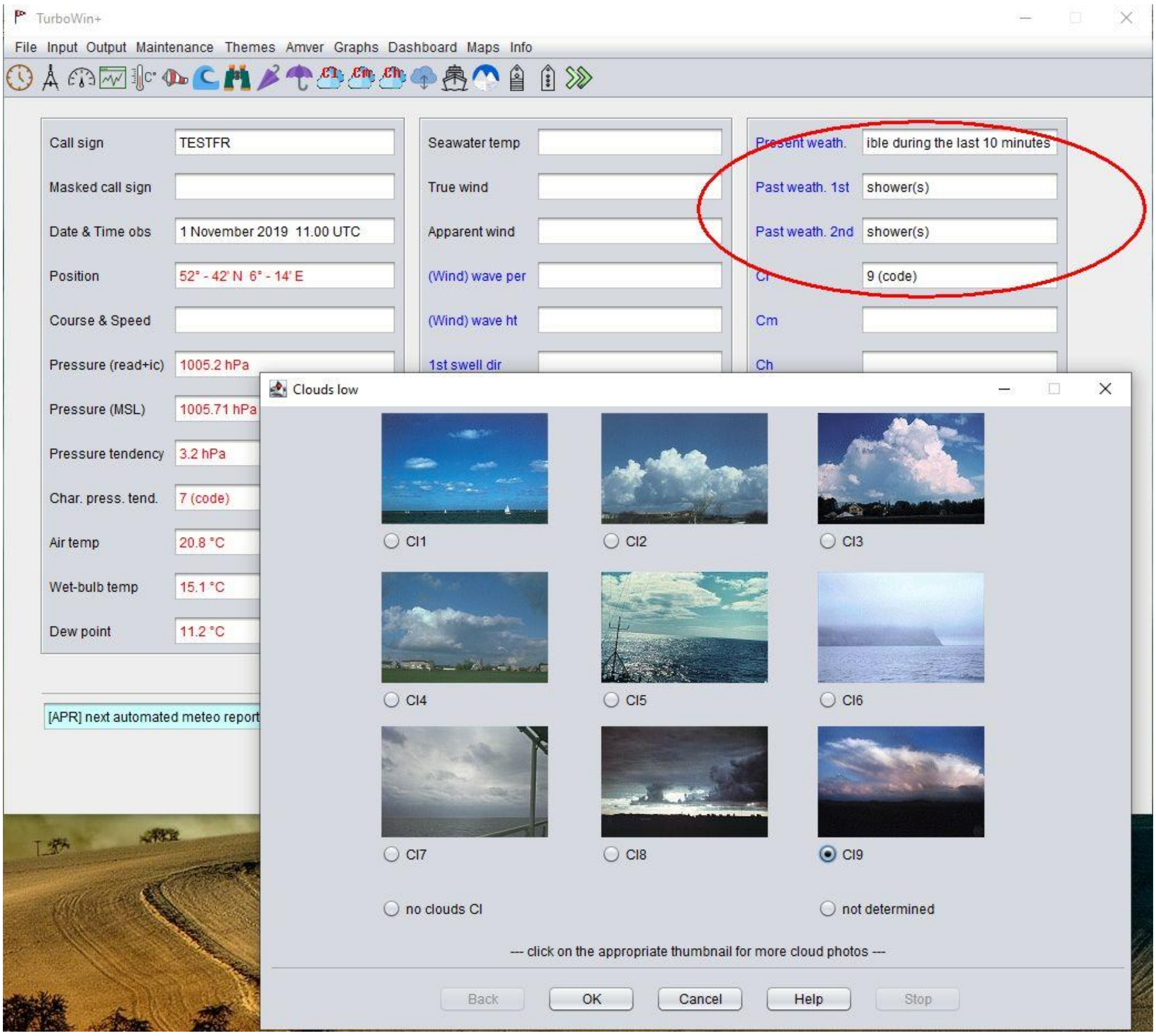


fig 21 adding manual observed data in APR/APTR mode

4.3 Switching between automatic modes APR/APTR and semi automatic mode

When in APR/APTR mode uncheck the AP[&T]R thickbox "report pressure [& temp] auto." and click OK (Fig 22). Not necessary to restart the computer. Now all output options and input screens are available as in the full manual mode or semi automatic mode. Check the thickbox again to continue with APR/APTR. Note, you also can switch APR on and off with the buttons on the main screen (see paragraph 4.4)

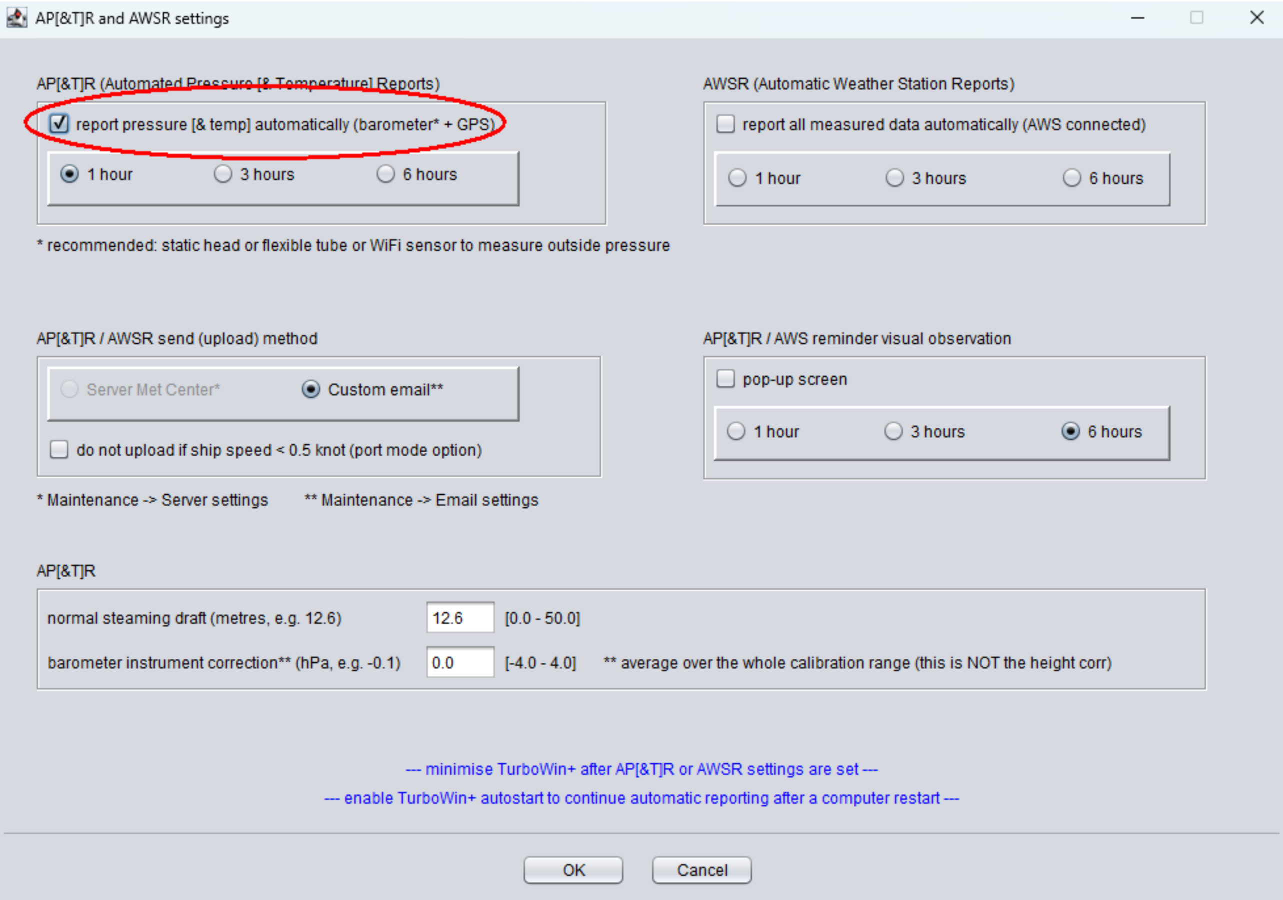


fig 22 switching between APR (APTR) and manual mode on settings screen

4.4 AP[&T]R and AWSR buttons on main screen

If the necessary setting were once made before (Maintenance → APR/APTR/AWSR settings). You also can turn on and off the AP[&T]R mode with one mouse click (Fig 23). Same for the AWSR mode.

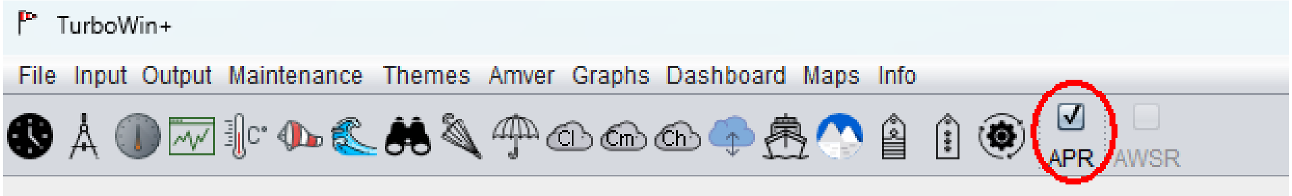


fig 23 switching between APR (APTR) and manual mode on main screen

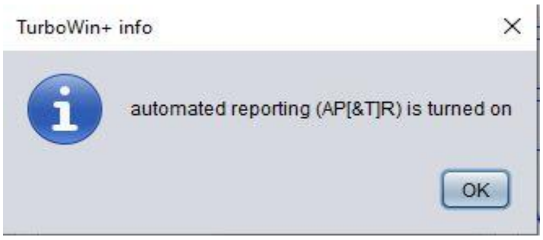


fig 24 APR (APTR) turned on message

5 Setting TurboWin+ to AWSR

Two options AWSR (Automated Weather Station Reports)

- EUCAWS device connected
- OMC-140 device connected

5.1 settings AWSR EUCAWS

- Select the observation format (Maintenance -> Obs format setting, Fig 25)
- Prepare the output send/upload options (Maintenance -> Email settings or Maintenance → Server settings)
- Select the EUCAWS device (Maintenance -> Serial/USB/LAN device settings, Fig 26)
- Set the AWSR options (Maintenance -> APR/APTR/AWSR settings, Fig 27)
- Optional: set the visual reminder pop-up screen (Fig 20b and Fig 20c)

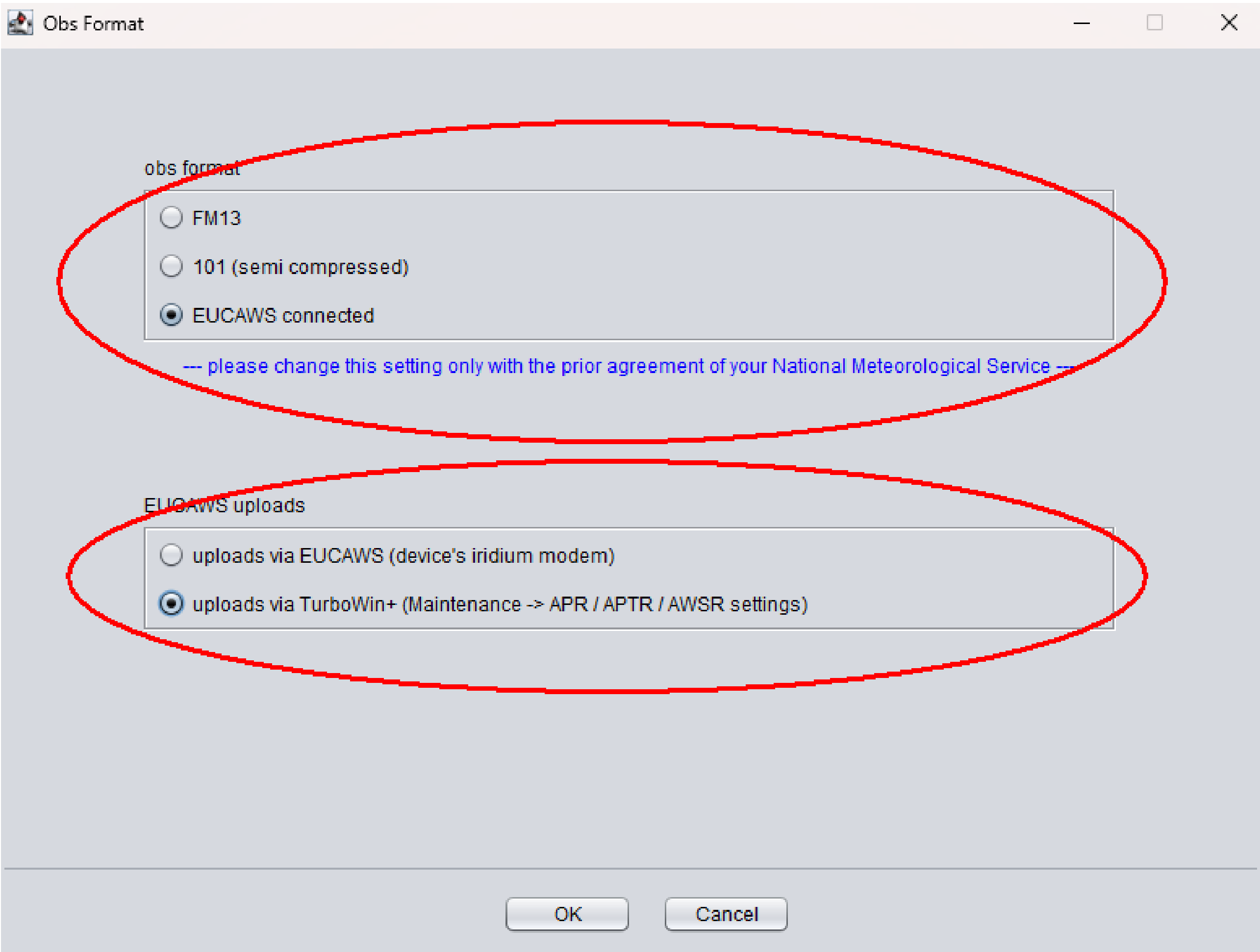


Fig 25 selecting obs format EUCAWS + uploads via TurboWin+ (as alternative for uploads via device's iridium modem)

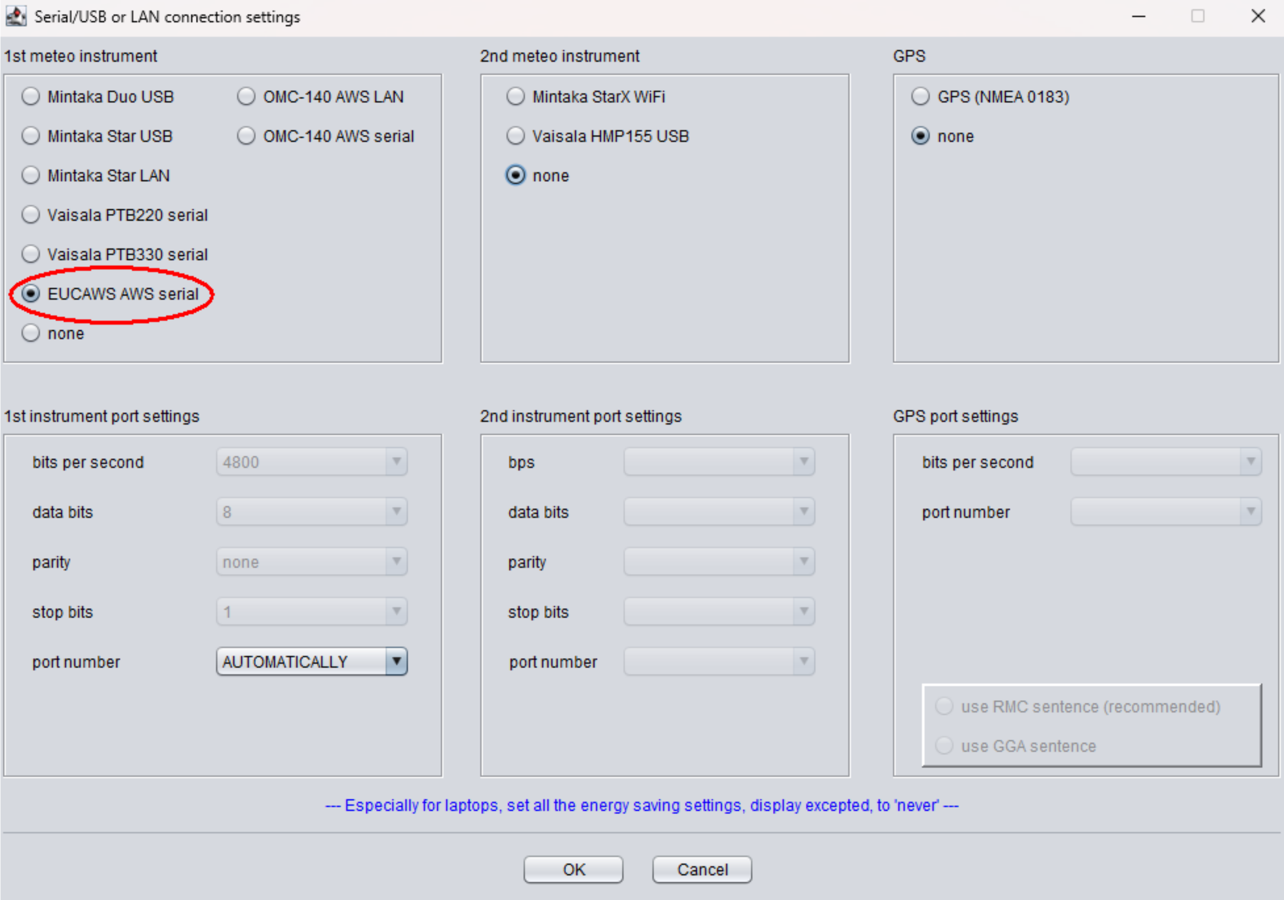


Fig 26 selecting the EUCAWS as connected device

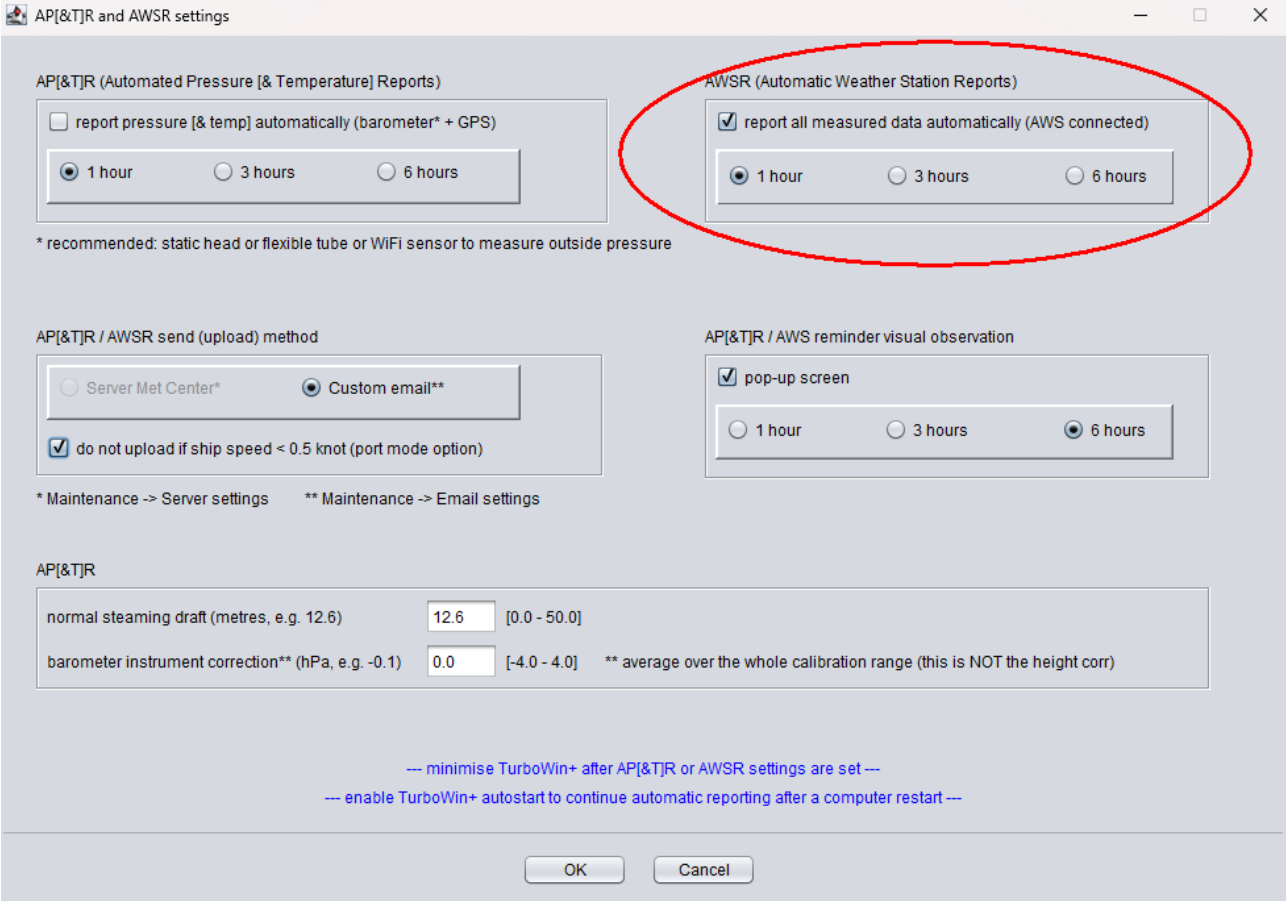


Fig 27 setting the AWSR options

5.2 settings AWSR OMC-140

- Select the 101 observation format (Maintenance -> Obs format setting, Fig 28)
- Prepare the output send/upload options (Maintenance -> Email settings or Maintenance → Server settings)
- Select the OMC-140 device (Maintenance -> Serial/USB/LAN device settings, Fig 29)
- Set the AWSR options (Maintenance -> APR/APTR/AWSR settings, Fig 27)
- Optional: set the visual reminder pop-up screen (Fig 20b and Fig 20c)

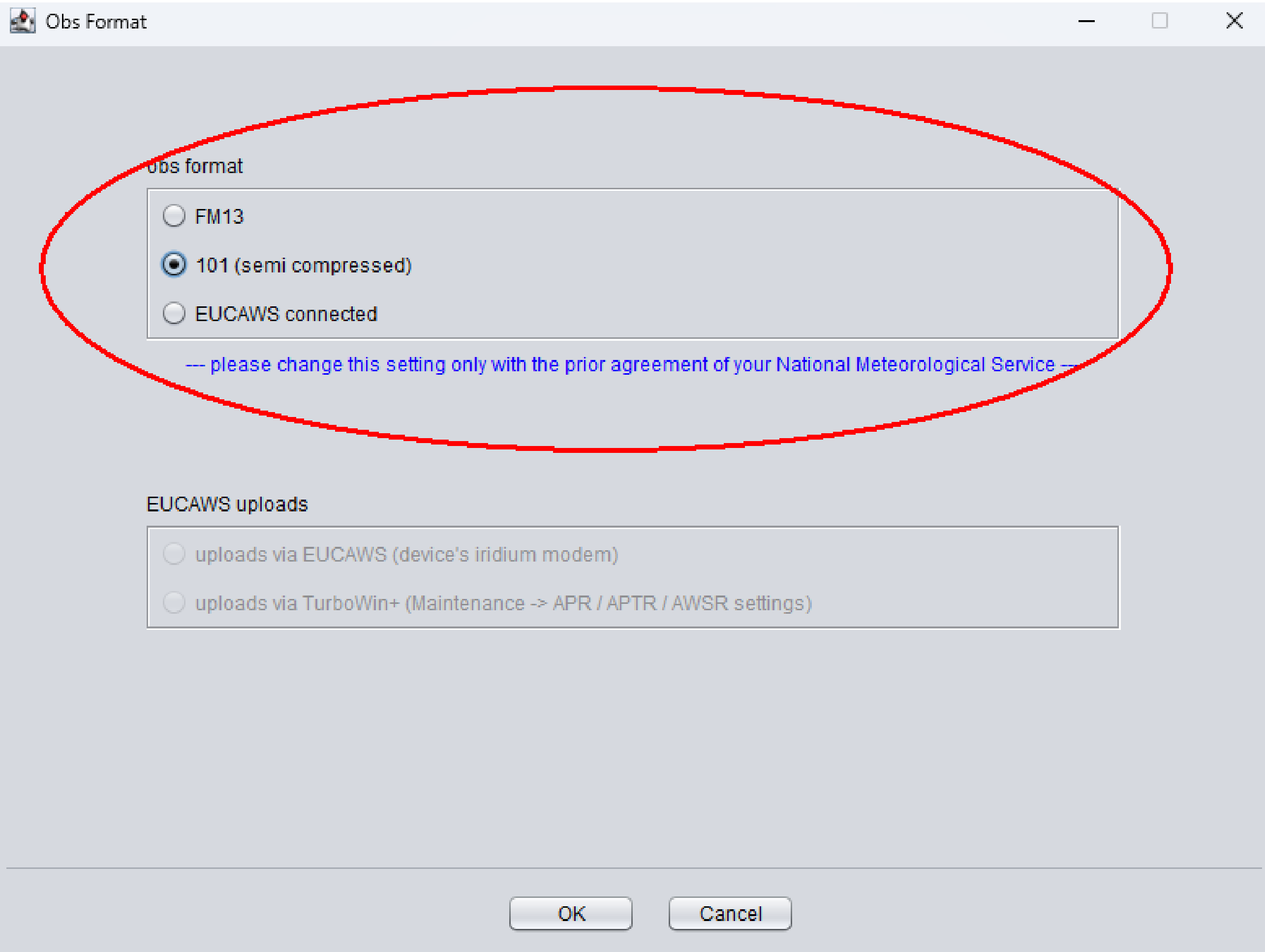


Fig 28 selecting obs format 101

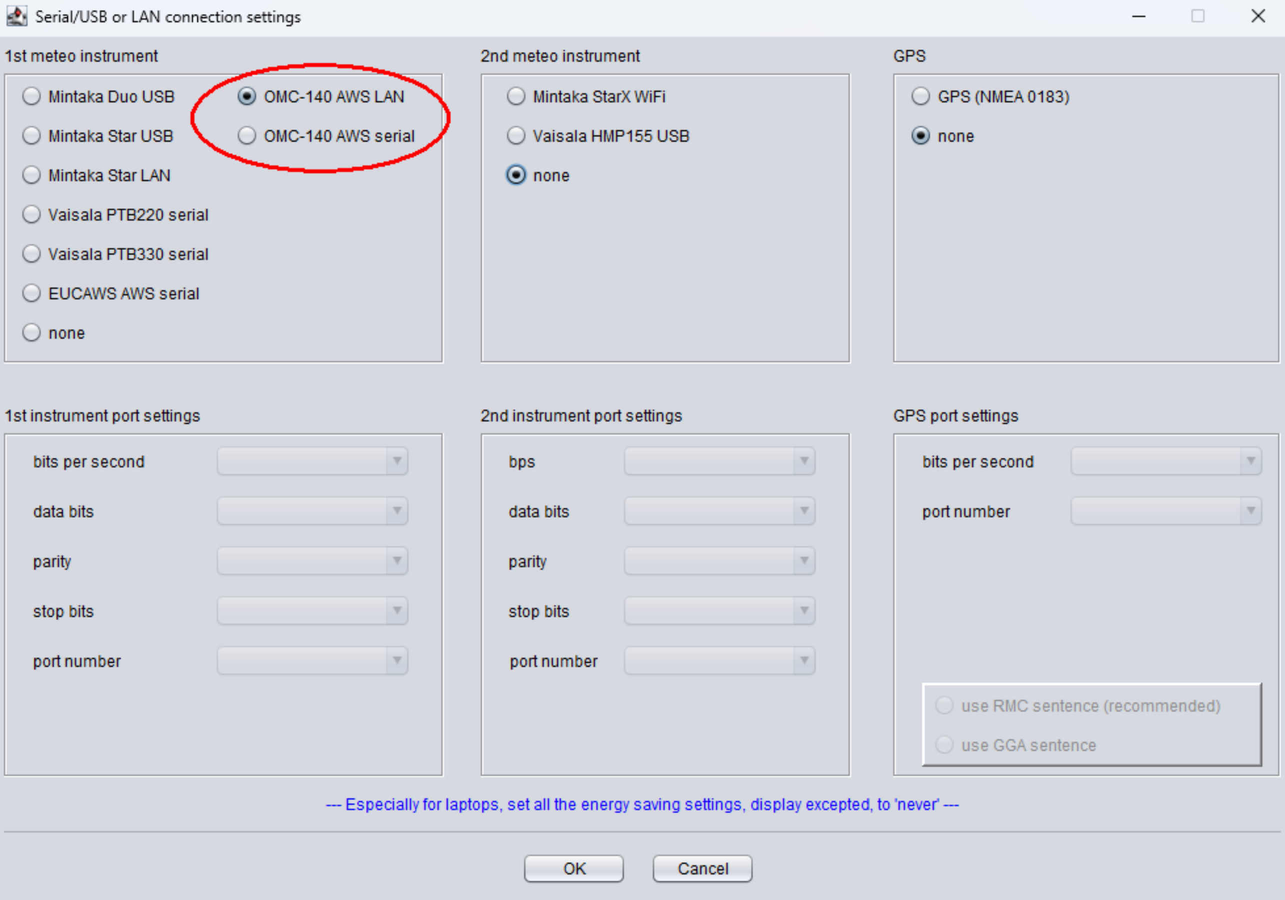


Fig 29 Selecting the OMC-140 device

5.3 Adding manual observed data when in AWSR mode

This is very easy. Open the appropriate meteorological parameter input screen, add the data and click OK (Fig 30). No further actions required. The inserted data will be automatically added to the observation with the next scheduled upload.

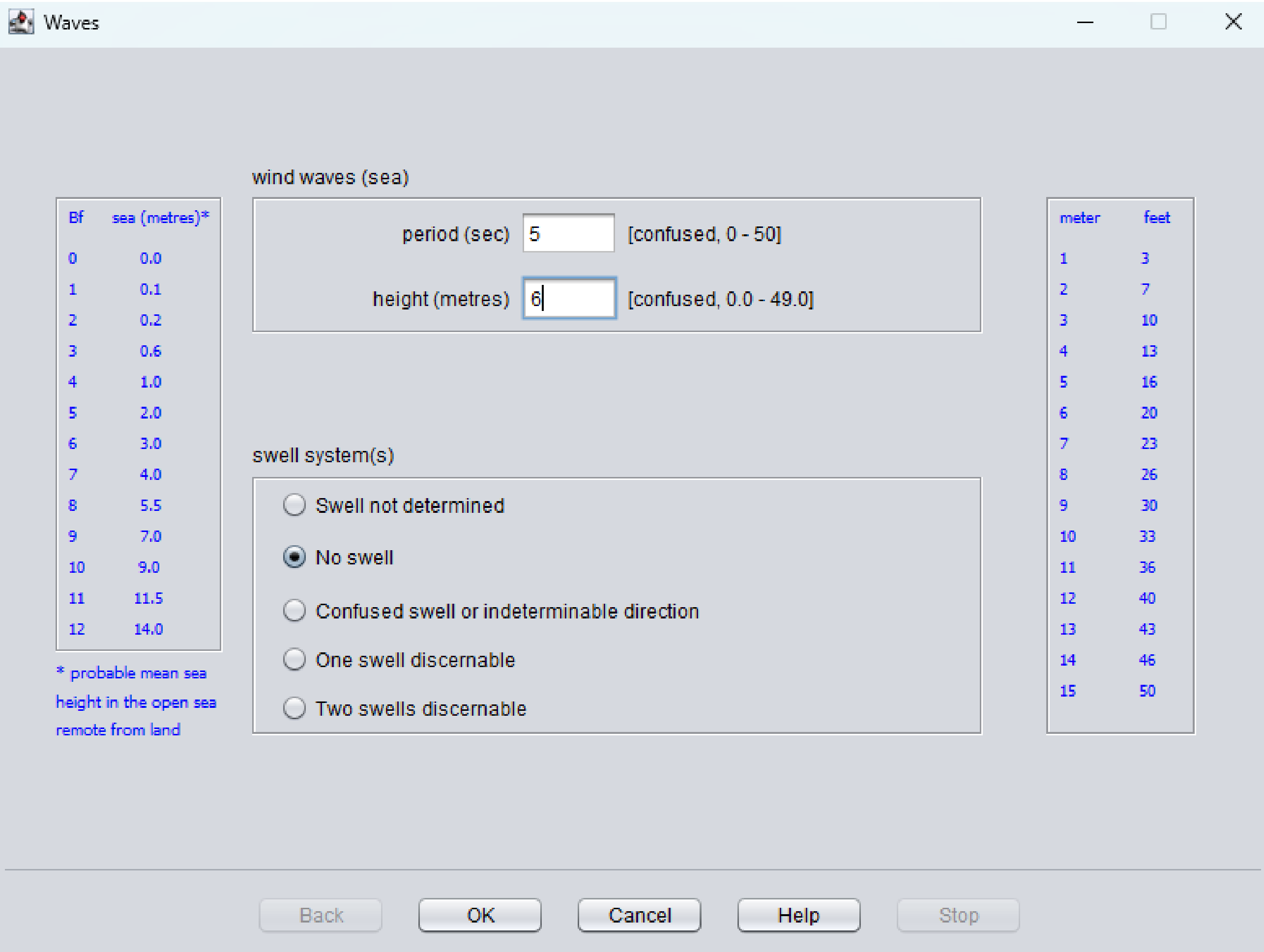


Fig 30 example, adding manual observed data in AWSR mode

6 Station ID

It was decided that there will be unique identifiers for marine platforms and instrument systems / packages on ships

See Fig 28 for the TurboWin+ station data input screen with the specific station ID input filed in red

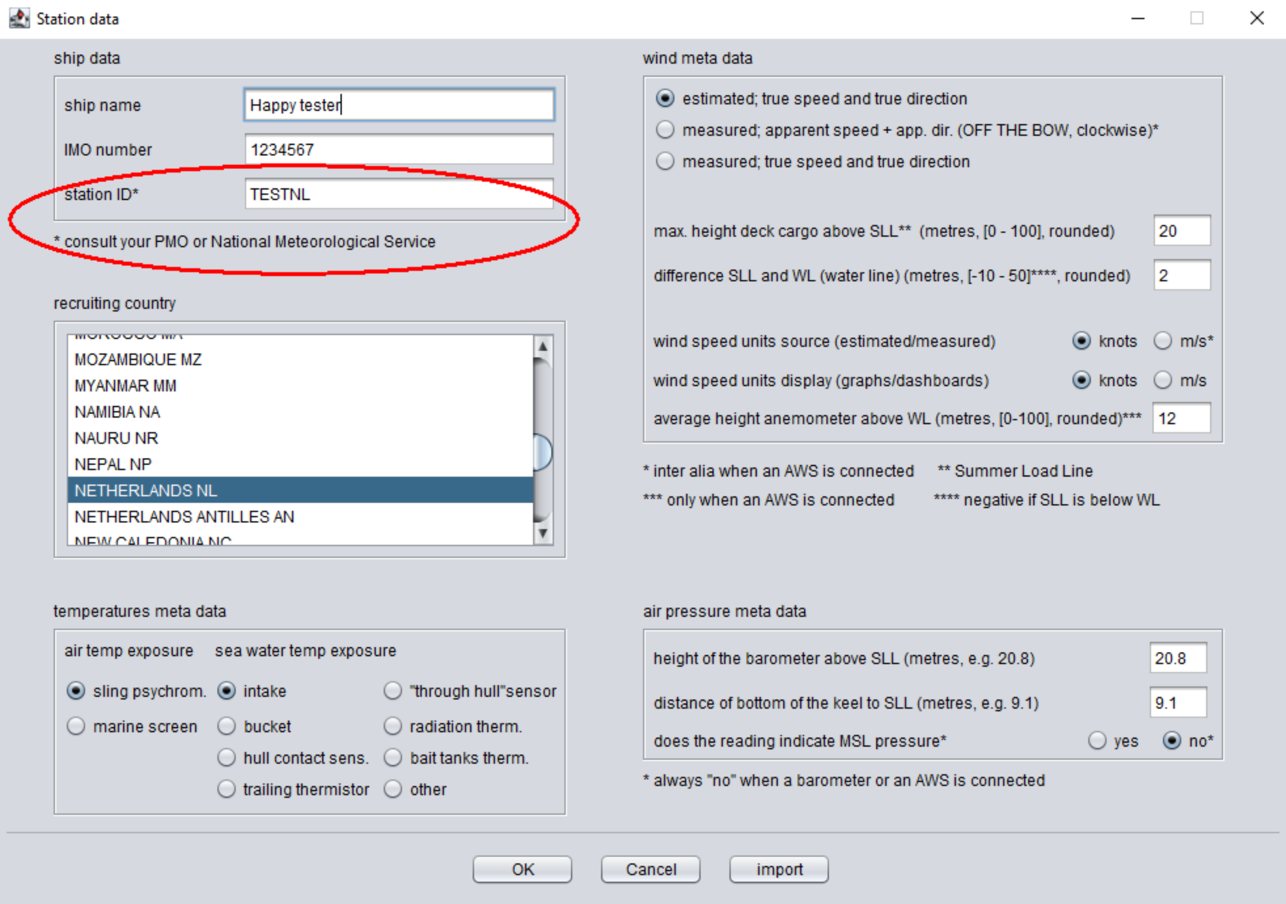


fig 28 Station data input screen

7 Dashboards

Dashboards available in AWS (Automatic Weather Station) connected mode and in APR/APTR (Automated Pressure [&Temperature] Reports) mode

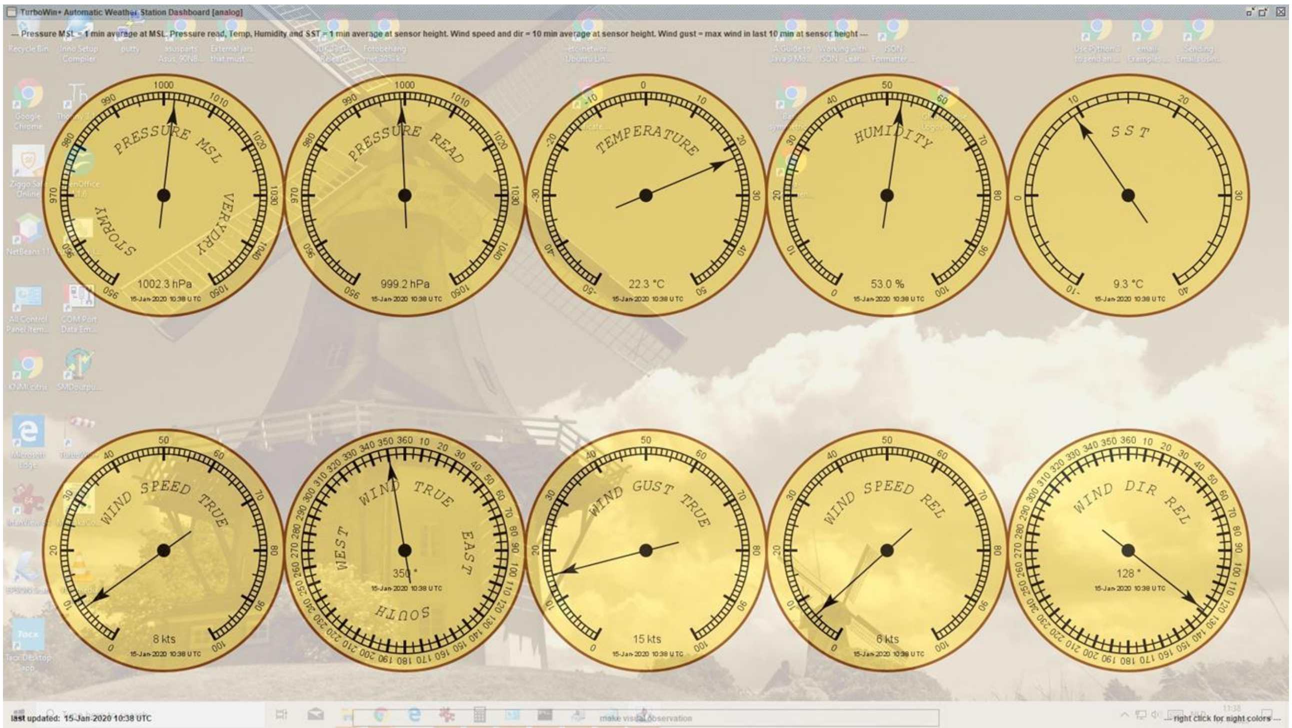


fig 30 AWS analog dashboard (in transparent screen mode)

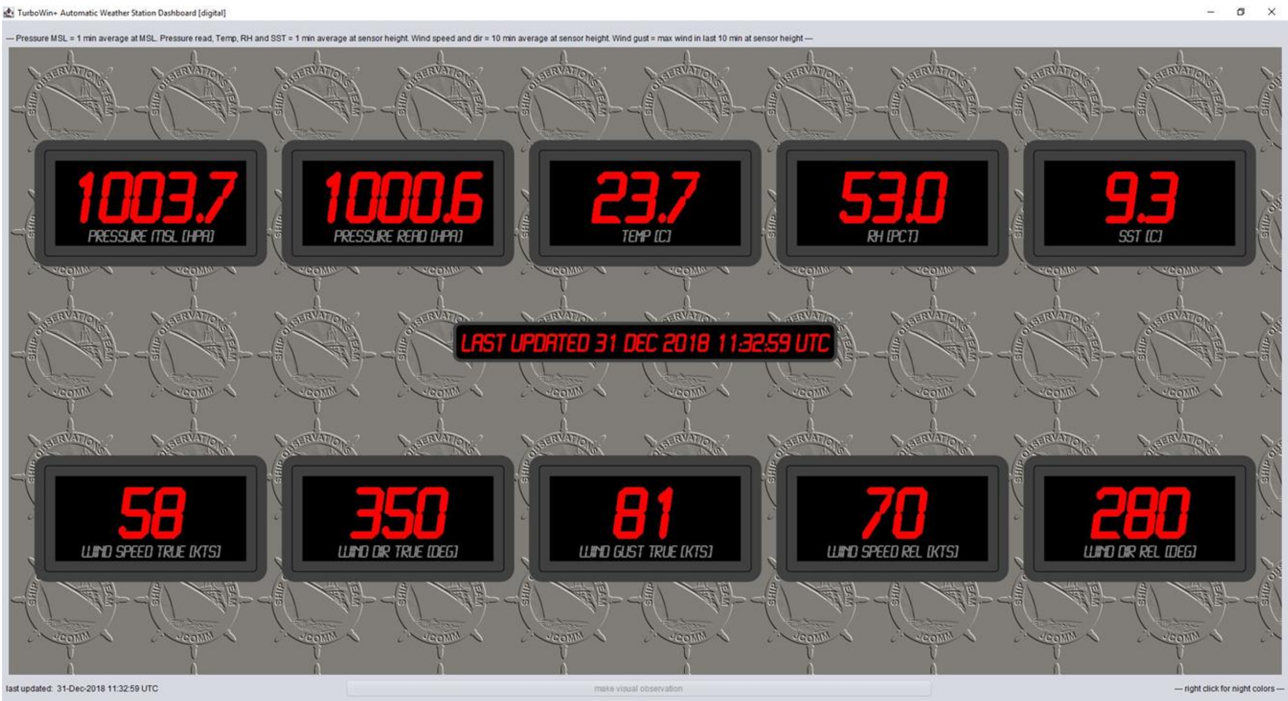


fig 31 AWS digital dashboard

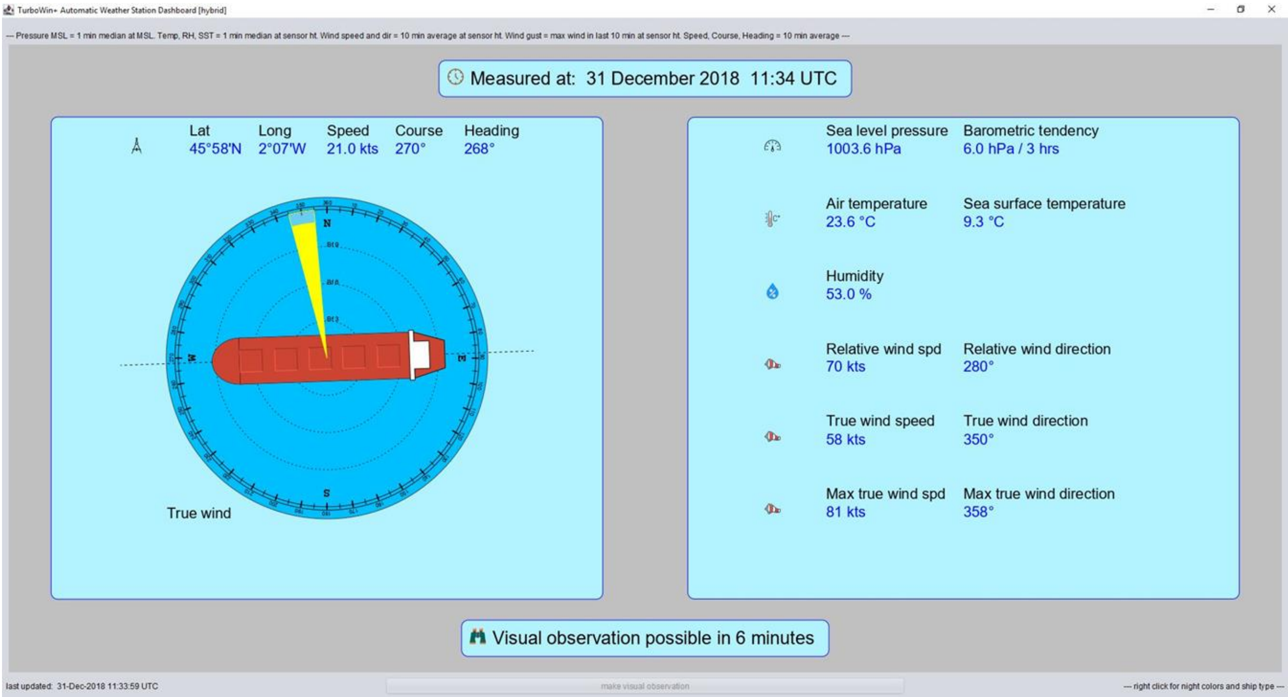


fig 32 AWS hybrid dashboard

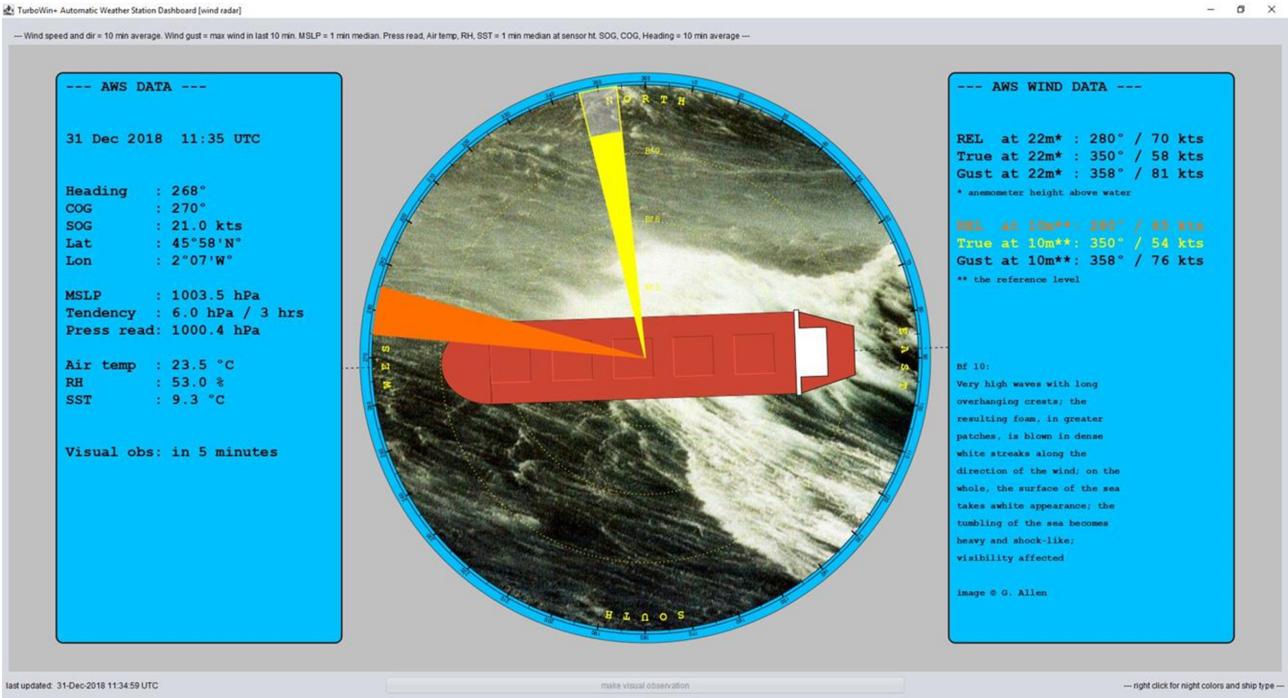


fig 33 AWS wind radar dashboard

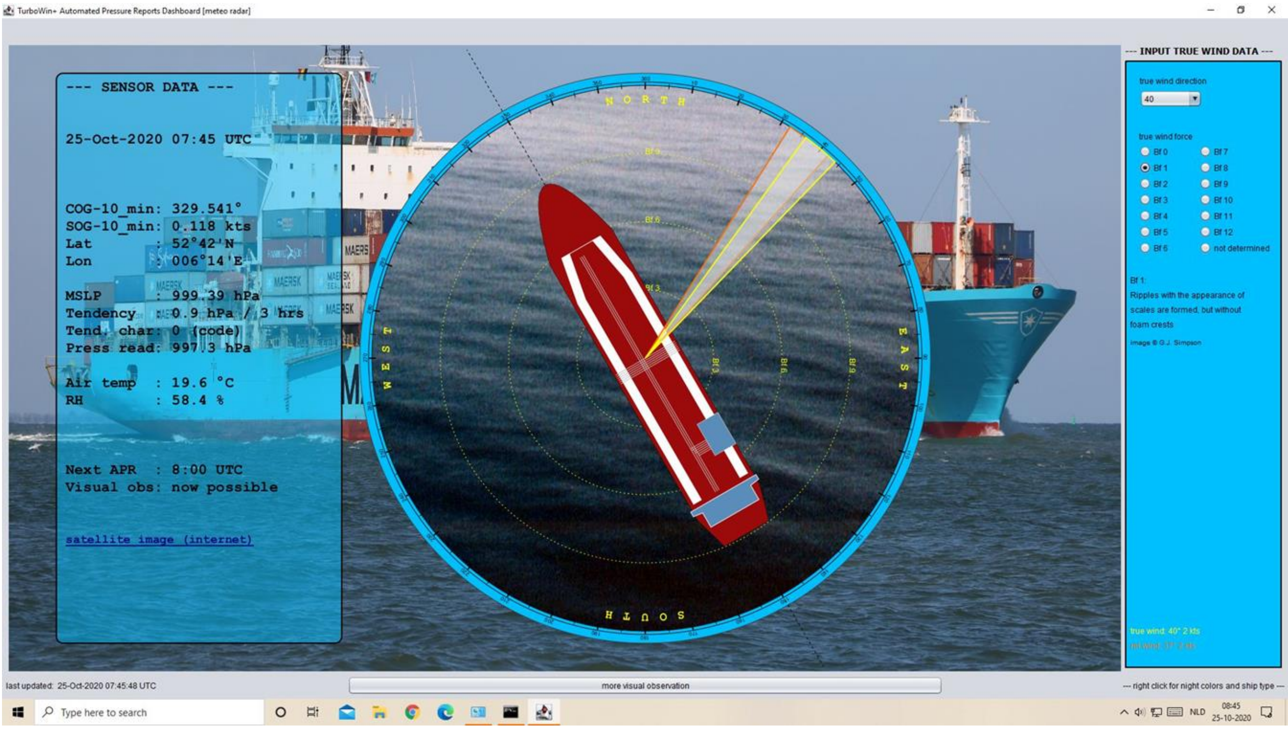


fig 34 APR meteo radar dashboard (day colours)

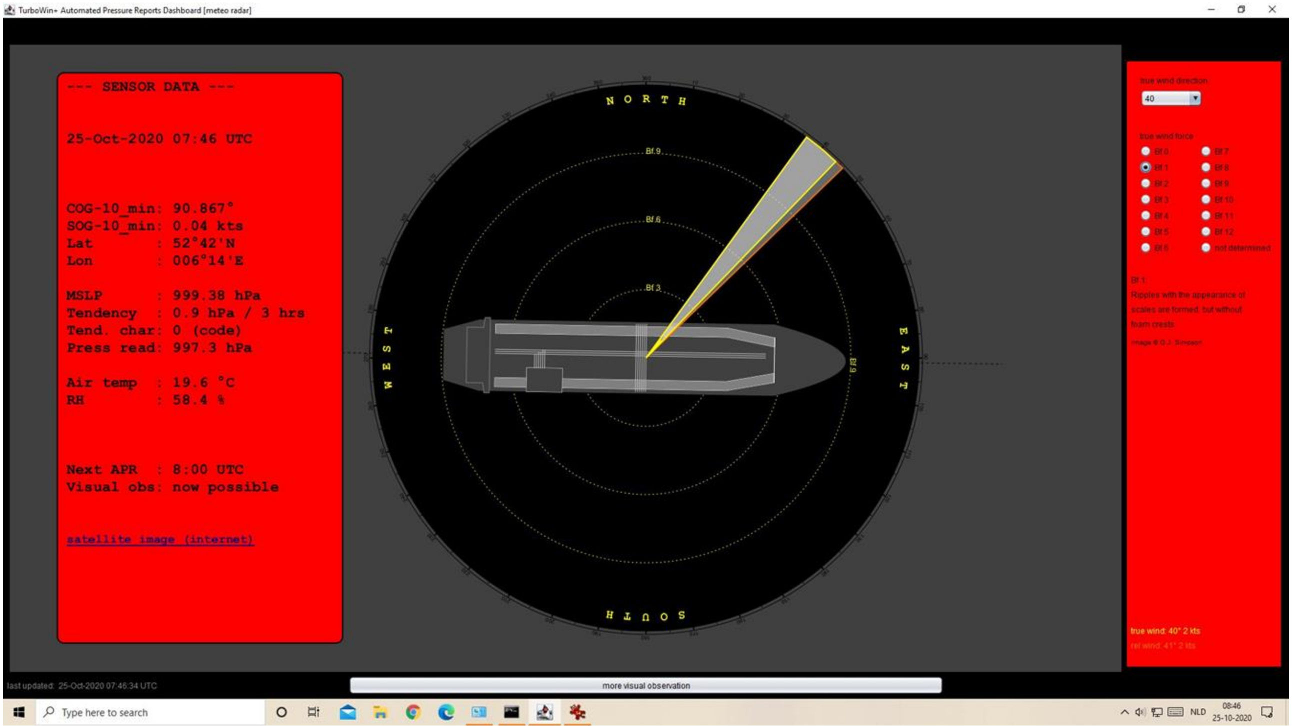


fig 35 APR meteo dashboard (night colours)

Selectable ship types in dashboards:

fig 36 dashboard ship types

8 Barometer comparison form (barometer self-check form)

A barometer comparison form was added. Select: Info → barometer comparison

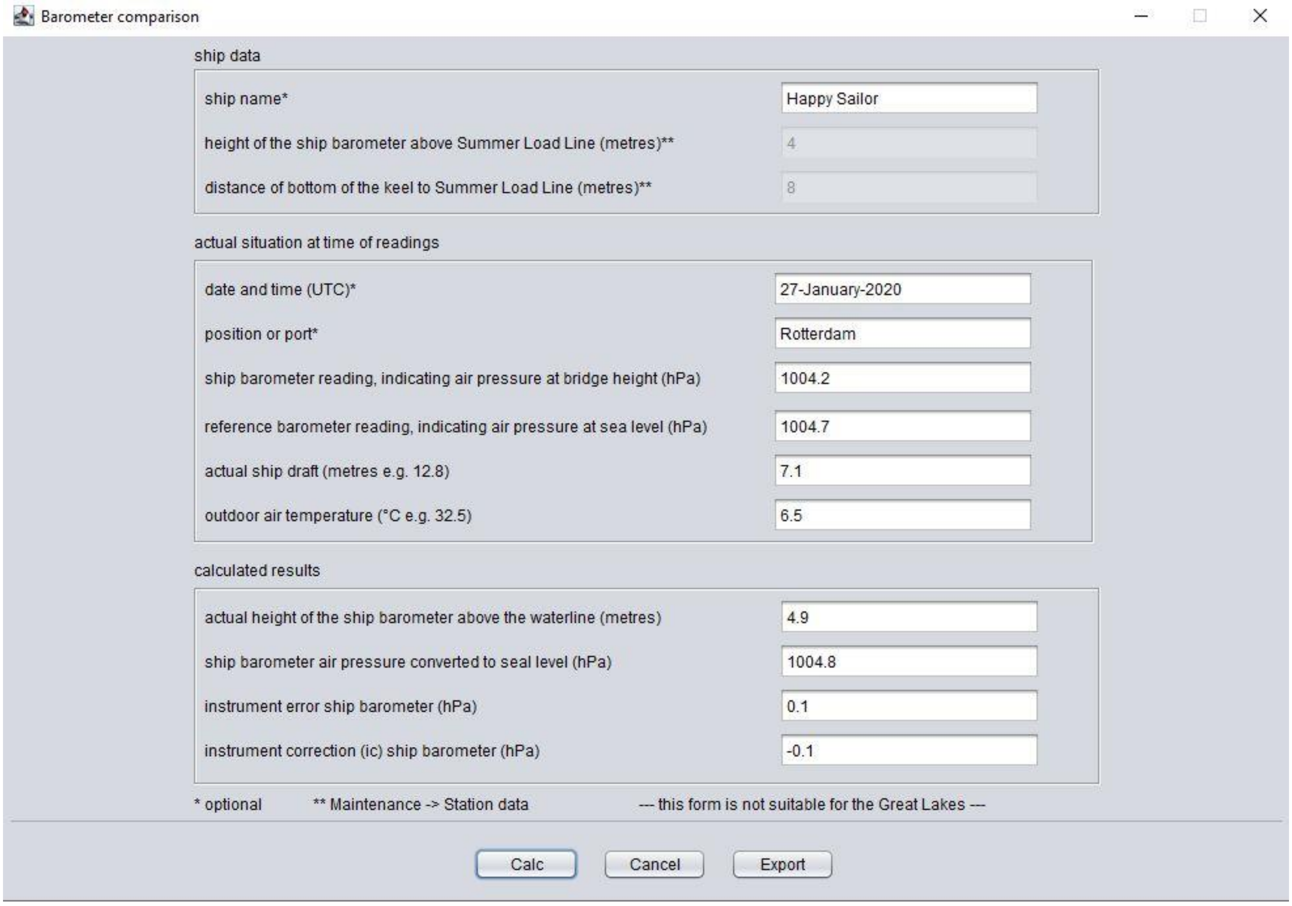


fig 41 barometer comparison form

9 Open source

TurboWin+ is free software and open-source software (https://www.gnu.org/licenses/gpl-3.0.en.html)

This allow users to run the software for any purpose as well as to study, change, and distribute it and any adapted versions. Anyone is freely licensed to use, copy, study, and change the software in any way, and the source code is openly shared on GitLab. (GPLv3 open-source license)

To emphasis the open source nature of TurboWin+ the official GNUv3 logo was added to the Info → About box.

Permissions:

- commercial use
- modification
- distribution
- patent use and private use

limitations:

- liability
- warranty

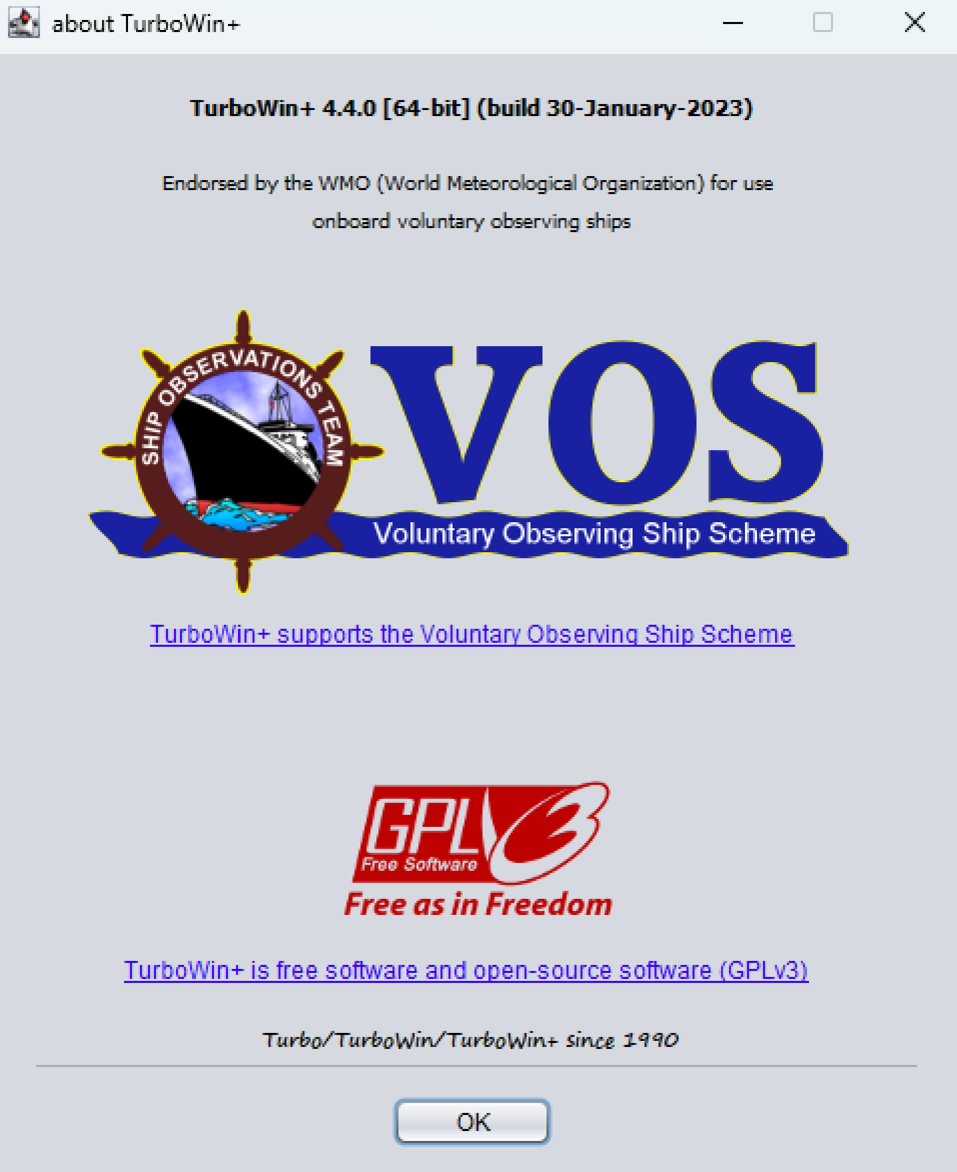
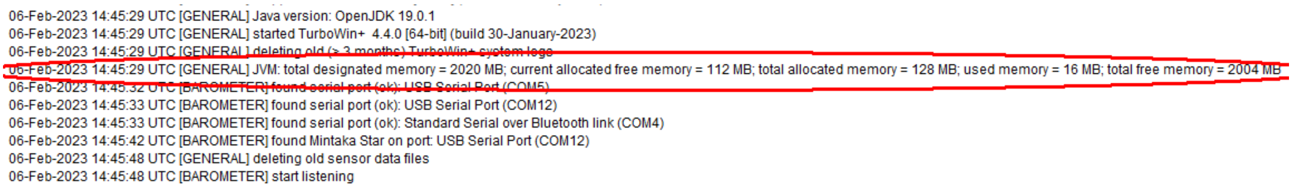


fig 42 About TurboWin+

10 Expert Tips

10.1) Memory check

At start up the TurboWin+ memory load of the pc was written to the TurboWin+ system log (Info → System log).



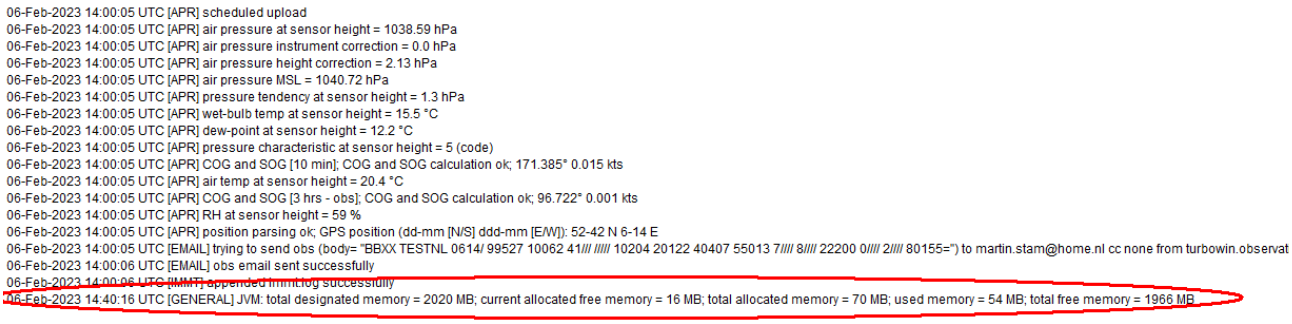
06-Feb-2023 14:45:29 UTC [GENERAL] Java version: OpenJDK 19.0.1
06-Feb-2023 14:45:29 UTC [GENERAL] started TurboWin+ 4.4.0 [64-bit] (build 30-January-2023)
06-Feb-2023 14:45:29 UTC [GENERAL] deleting old (> 3 months) TurboWin+ system logs
06-Feb-2023 14:45:29 UTC [GENERAL] JVM: total designated memory = 2020 MB; current allocated free memory = 112 MB; total allocated memory = 128 MB; used memory = 16 MB; total free memory = 2004 MB
06-Feb-2023 14:45:32 UTC [BAROMETER] found serial port (ok): USB Serial Port (COM5)
06-Feb-2023 14:45:33 UTC [BAROMETER] found serial port (ok): USB Serial Port (COM12)
06-Feb-2023 14:45:33 UTC [BAROMETER] found serial port (ok): Standard Serial over Bluetooth link (COM4)
06-Feb-2023 14:45:42 UTC [BAROMETER] found Mintaka Star on port: USB Serial Port (COM12)
06-Feb-2023 14:45:48 UTC [GENERAL] deleting old sensor data files
06-Feb-2023 14:45:48 UTC [BAROMETER] start listening

fig 43 example memory load TurboWin+ at start-up

You can also check the memory load when TurboWin+ is running (snapshot) by clicking on one of the application exit options <u>but not confirming the exit</u> (click on the NO button), see fig 44

fig 44 TurboWin+ Exit confirmation message

After clicking NO, select Info -> System log. A memory load line was again written in the log, see fig 45



06-Feb-2023 14:00:05 UTC [APR] scheduled upload
06-Feb-2023 14:00:05 UTC [APR] air pressure at sensor height = 1038.59 hPa
06-Feb-2023 14:00:05 UTC [APR] air pressure instrument correction = 0.0 hPa
06-Feb-2023 14:00:05 UTC [APR] air pressure height correction = 2.13 hPa
06-Feb-2023 14:00:05 UTC [APR] air pressure MSL = 1040.72 hPa
06-Feb-2023 14:00:05 UTC [APR] pressure tendency at sensor height = 1.3 hPa
06-Feb-2023 14:00:05 UTC [APR] wet-bulb temp at sensor height = 15.5 °C
06-Feb-2023 14:00:05 UTC [APR] dew-point at sensor height = 12.2 °C
06-Feb-2023 14:00:05 UTC [APR] pressure characteristic at sensor height = 5 (code)
06-Feb-2023 14:00:05 UTC [APR] COG and SOG [10 min]; COG and SOG calculation ok; 171.385° 0.015 kts
06-Feb-2023 14:00:05 UTC [APR] air temp at sensor height = 20.4 °C
06-Feb-2023 14:00:05 UTC [APR] COG and SOG [3 hrs - obs]; COG and SOG calculation ok; 96.722° 0.001 kts
06-Feb-2023 14:00:05 UTC [APR] RH at sensor height = 59 %
06-Feb-2023 14:00:05 UTC [APR] position parsing ok; GPS position (dd-mm [N/S] ddd-mm [E/W]): 52-42 N 6-14 E
06-Feb-2023 14:00:05 UTC [EMAIL] trying to send obs (body= "BBXX TESTNL 0614/ 99527 10062 41/// ///// 10204 20122 40407 55013 7//// 8//// 22200 0//// 2//// 80155=") to martin.stam@home.nl cc none from turbowin.observat
06-Feb-2023 14:00:06 UTC [EMAIL] obs email sent successfully
06-Feb-2023 14:00:06 UTC [IMMT] appended immt.log successfully
06-Feb-2023 14:40:16 UTC [GENERAL] JVM: total designated memory = 2020 MB; current allocated free memory = 16 MB; total allocated memory = 70 MB; used memory = 54 MB; total free memory = 1966 MB

fig 45 example snapshot memory load TurboWin+

10.2) COM ports (Windows)

“I*n case you are facing with trouble with your COM ports in use that stacking, these brief steps below will help you to clear them out (for example, when you found myself needing to connect via a USB to Serial adapter to a server and noticed that 35 COM ports showed in use). Somehow the COM ports were not getting cleared out.*

The fact is that when you connect a new COM or a USB device (such as smartphone, modem, RS232 to USB converters, etc.), Windows detects newly connected device using Plug-and-Play (PnP) and creates a new virtual COM port with a number from 1 to 255. When you remove this device, the created COM port will not be deleted automatically. If you reconnect this device to a computer, the system re-assigned to it the reserved COM port number. But for any new COM device is assigned the first unallocated (free) COM port number. Thus, over time, a large number of reserved COM ports (in “in use” state) for certain COM or USB devices may appear in the system, although the devices themselves may not be connected to your computer. These ports by-default are not visible to the user in Device Manager”. Continue reading : https://theitbros.com/how-to-delete-com-ports-in-use/

10.3) USB hub

If you use a USB hub it is highly recommended to use a powered hub to avoid potential communication errors. Powered hubs (also called active hubs) come with their own AC adapter and plug into the mains (Fig 46)

fig 46 powered USB hub

10.4) Mintaka Star Bridged connection

If the pc onboard is connected with a Ethernet cable to the internet you can connect the Mintaka Star with an USB cable to the pc. But, as an alternative, you can also use a network bridge by setting the Mintaka Star to WiFi mode (Access Point mode, Star creates its own network) and creating a network bridge under Windows in the Network Connections section (Fig 47)

Note

After every pc restart you have to reconnect the bridge!

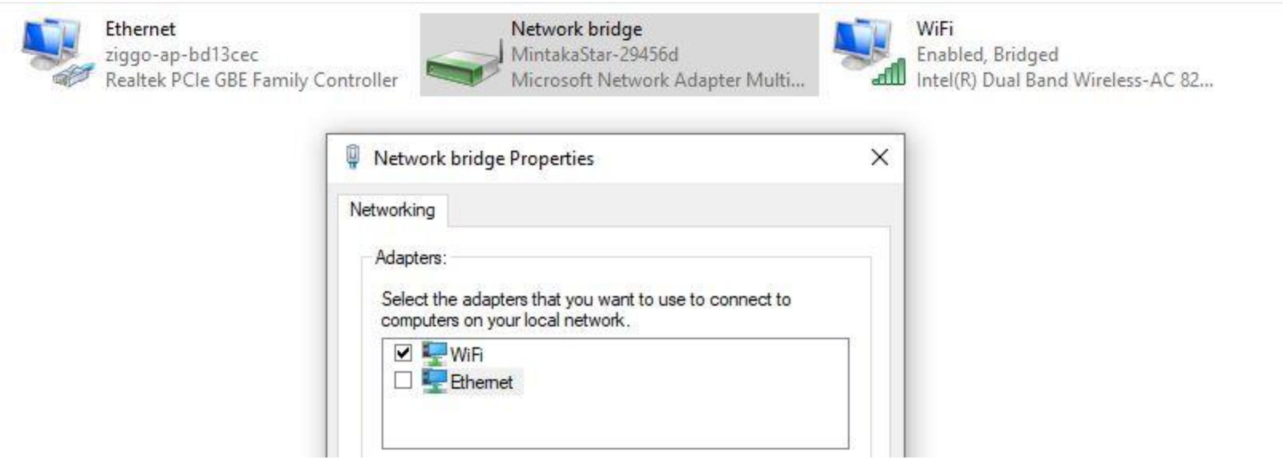


Fig 47 Windows control panel network connections; Mintaka Star via a network bridge

10.5) Verbose mode

For test purposes it is possible to start TurboWin+ in verbose mode by means of invoking "turbowin_launcher.bat"

(e.g. Windows 64 bit: Full path C:\Program Files (x86)\TurboWin+\bin\turbowin_launcher.bat). A console window will be opened (Fig 48).

note

Do not use this option in operational mode

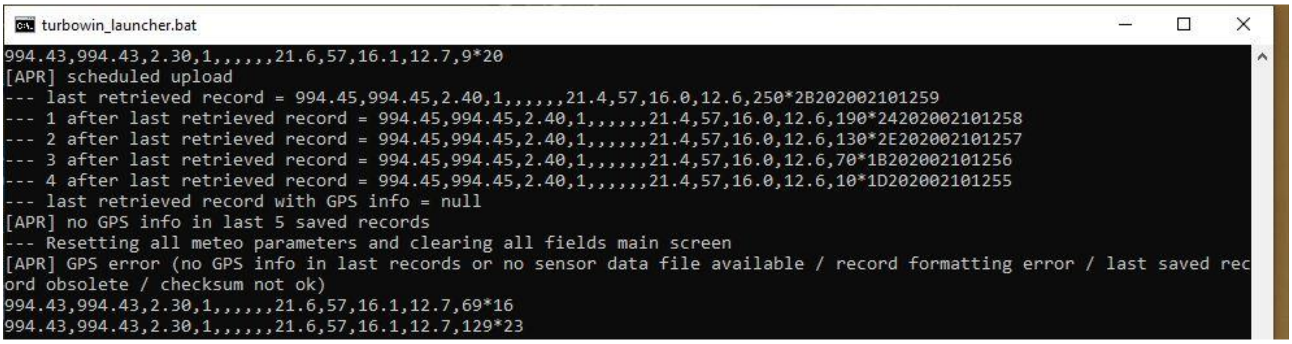


fig 48 TurboWin+ console window

10.6) Check more than one instance of TurboWin+ (Windows OS)

TurboWin+ is based on Java. It is not easy for Java programs to check that not more than one instance of an application is running. A well-known and generally accepted approach in the Java community is to claim a never used network computer port (e.g. 12345). So a second instance of TurboWin+ is trying to do the same but get a message that the network port is already occupied (Fig 49)

fig 49 TurboWin+ was invoked but was already running

Now is it theoretically possible that another Java program is doing the same, or port 12345 is closed by security policy. In the latter case please add port 12345 to the firewall exception list, in the first case define a different port number at TurboWin+ start up by right clicking on the TurboWin+ desktop icon and add a (random) different port number (see fig 50a).

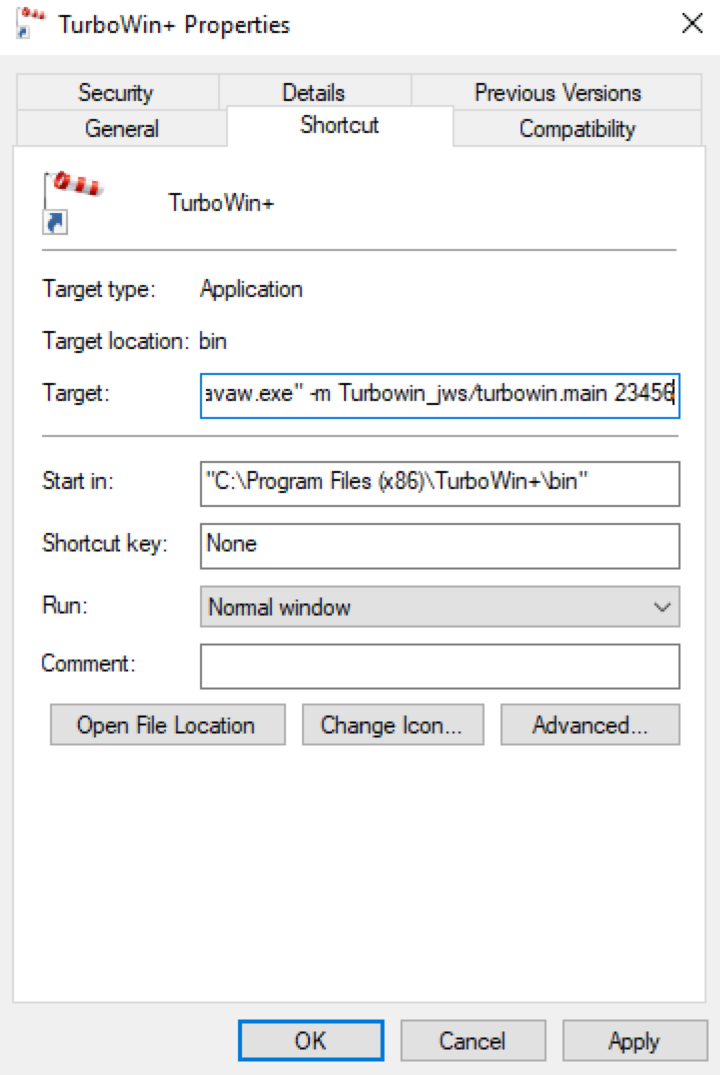


Fig 50a example desktop icon shortcut with network port 23456 as alternative check multiple instances running

In verbose mode, for testing, it is also possible to start TurboWin+ with another random (multiple-instances-check) network port. This can be done by creating a shortcut to “turbowin_launcher.bat” and adding the new network port number on the target line (Fig 50b).

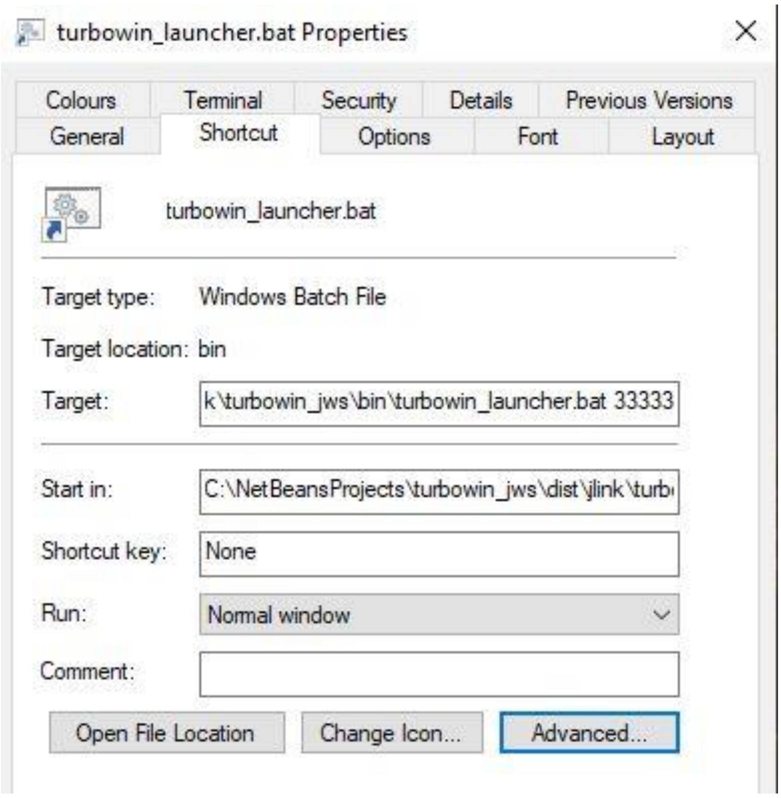


fig 50b TurboWin+ example launcher shortcut with network port 33333 as alternative check multiple instances running

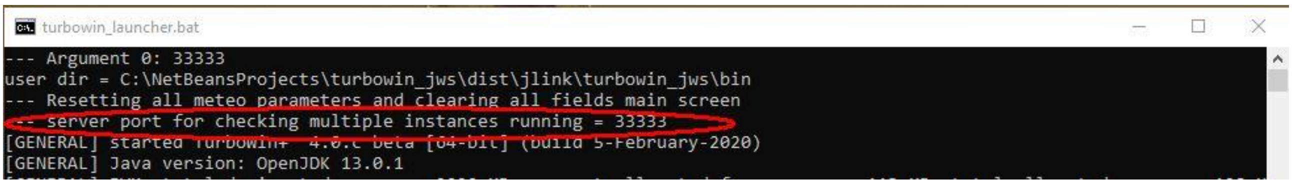


fig 51 TurboWin+ console window, network port 33333 used for checking more than one TurboWin+ instance is running

The console window will show the result, see Fig 51.

note

Do not use this (invoking TurboWin+ via “turbowin_launcher.bat”) option, in operational mode. In operational mode use the desktop icon shortcut update option (see above, fig 50a)

10.7) Email log

Besides the default logging in the TurboWin+ system log (Info → System log) you can also check the dedicated email log if the mail was sent with the secondary Custom email option (Maintenance → Email settings). Location of this log: ../logs/python/log_python_email.txt. See Fig 52

log_python_email.txt - Notepad

File Edit Format View Help

```
[date-time logging:  05 Jul 2021 11:00:19  UTC ]
TurboWin+ python email module (build 3-February-2021 13:00 UTC)
Script name:  C:\Program Files (x86)\TurboWin+\bin\logs\python\email_tbw.exe
Number of arguments:  11
The arguments are:
smtp_mode           =  CUSTOM_TLS_STARTTLS
smtp_host_local     =  smtp.mail.yahoo.com
smtp_password_local =  ******
send_to             =  martin.stam@home.nl
send_from           =  turbowin@yahoo.com
subject             =  obs TEST vivobook
body                =  BBXX TESTNL 0511/ 99427 10160 41/// ///// 10234 20149 40052 58002 7//// 8//// 222// 0//// 2//// 80180=
send_cc             =  null
smtp_port_local     =  587
attachment          =  null

mail sent successfully to:  martin.stam@home.nl
```

fig 52 dedicated email log (if secondary email module is used)

10.8) Format #101 logging

There are dedicated format#101 generated logs (although partly in the French language, they could be helpful when trouble shooting) available after one of the output options in format#101 mode was invoked. These logs are always generated in format #101 mode.

Locations:

../logs/format_101/log/log_compression.txt (Fig 43)

../logs/format_101/log/log_decompression.txt (Fig 44)

log_compression.txt - Notepad

File Edit Format View Help

```
ficin=temp/format_101.txt
REP= temp/
 indicatif=3SD234X


TEST mode operatoire : 0
i=0 1 101.000000
i=1 1 0.000000
i=2 0 0.000000
i=3 0 0.000000
i=4 0 0.000000
i=5 0 0.000000
i=6 1 2020.000000
i=7 1 2.000000
i=8 1 3.000000
i=9 1 14.000000
i=10 1 0.000000
i=11 1 52.700000
i=12 1 6.233000
i=13 1 100970.000000
i=14 1 101030.000000
i=15 1 30.000000
```

fig 53 -part of- format#101 compression log

log_decompression.txt - Notepad

File Edit Format View Help

```
005002 -  - 52.700000 - Latitude (coarse accuracy)
006002 -  - 6.230000 - Longitude (coarse accuracy)
010004 -  - 100970.000000 - Pressure at barometer height
010051 -  - 101030.000000 - Pressure reduced to mean sea level
010061 -  - 30.000000 - 3-hour pressure change
010063 -  - 0.000000 - Characteristic of pressure tendency
011001 -  -   ---ABSENT--- - True wind direction clockwise from the north
011002 -  -   ---ABSENT--- - True wind speed
011099 -  -   ---ABSENT--- - Relative wind direction clockwise from the bow
011100 -  -   ---ABSENT--- - Relative wind speed
011041 -  -   ---ABSENT--- - Maximum wind gust speed
011043 -  -   ---ABSENT--- - Maximum wind gust direction
012101 -  - 294.250000 - Air temperature
012102 -  - 289.050000 - Wet bulb temperature
012103 -  - 285.650000 - Dew point temperature
013003 -  - 58.400000 - Relative humidity
022042 -  -   ---ABSENT--- - Sea temperature
020001 - code -   ---ABSENT--- - Horizontal visibility
020003 -  -   ---ABSENT--- - Present weather
020004 -  -   ---ABSENT--- - Past weather 1
020005 -  -   ---ABSENT--- - Past weather 2
020011 - N -   ---ABSENT--- - Total cloud cover
020011 - Nh -   ---ABSENT--- - Cloud amount (low)
020012 - CL_100 -   ---ABSENT--- - Cloud type (low)
020012 - CM_100 -   ---ABSENT--- - Cloud type (middle)
020012 - CH_100 -   ---ABSENT--- - Cloud type (high)
020013 - code -   ---ABSENT--- - Height of base of lowest clouds
022012 -  -   ---ABSENT--- - Period of wind waves
022022 -  -   ---ABSENT--- - Height of wind waves
022003 - SW1 -   ---ABSENT--- - Direction of 1st swell
022013 - SW1 -   ---ABSENT--- - Period of 1st swell
022023 - SW1 -   ---ABSENT--- - Height of 1st swell
022003 - SW2 -   ---ABSENT--- - Direction of 2nd swell
022013 - SW2 -   ---ABSENT--- - Period of 2nd swell
022023 - SW2 -   ---ABSENT--- - Height of 2nd swell
----------------------------------------------------------------
004192;OBS;1;2020/02/03 14:00:00; <-- date-heure observation
ENTETE_SHIP:YYGGw= 03144
==================== CALCUL RETARD DE DIFFUSION ========================
date courante : heure=14 minute=0 secondes=19 jour=3 mois=2 annee=2020 temps=1580738419
date obs      : heure=14 jour=3 mois=2 annee=2020 temps=1580738400
LAPS          : 0.005278h
entete=SNVF25 LFPW 031400
```

fig 54 -part of- format#101 decompression log

More additional format#101 log data can be found in directory: ../logs/format_101/temp/

10.9) Linux (64bit) installation

RPM and DEB packages are available.

Note 1 USB

If an instrument will be connected to TurboWin+ you must have full and direct access to the USB ports, to set these permissions on Ubuntu (once):

sudo adduser <the user you want to add> dialout

sudo reboot

Note 2

For troubleshooting you can start TurboWin+ with the launcher to monitor all the incoming and outgoing messages,

go to the appropriate directory and start: *./turbowin_launcher*

10.10) Obs to server (Rest API)

<u>TurboWin+ Server-side API</u>

In TurboWin+ there are options to send (upload) a meteorological observation straight to the National Meteorological Service or to a Centralized Service. This is the fastest way to get an observation from the ship into the GTS. There are slight differences in the server-side API implementation of FM13 and format #101 observations.

<u>Format #101 API</u>

In TurboWin+ (from version 2.5.1) there is the option to send the format #101 observation straight to a server of a National Meteorological Service or Centralized Service.

<u>Upload URL format 101:</u>

TurboWin+ sends a GET HTTP request to the URL which can be set in TurboWin+.
The server must expect the HTTP request, in the form of GET. When received, the server must interpret and validate the information supplied and respond as below

upload URL which can be set manually as a setting in TurboWin+: --- consult your PMO ---

So the server site script could be placed on every server with every path and could be named "index_webstart_101.php" but also this is not mandatory. But of course the full URL must be set once manually in TurboWin+ and must always end with a "?"

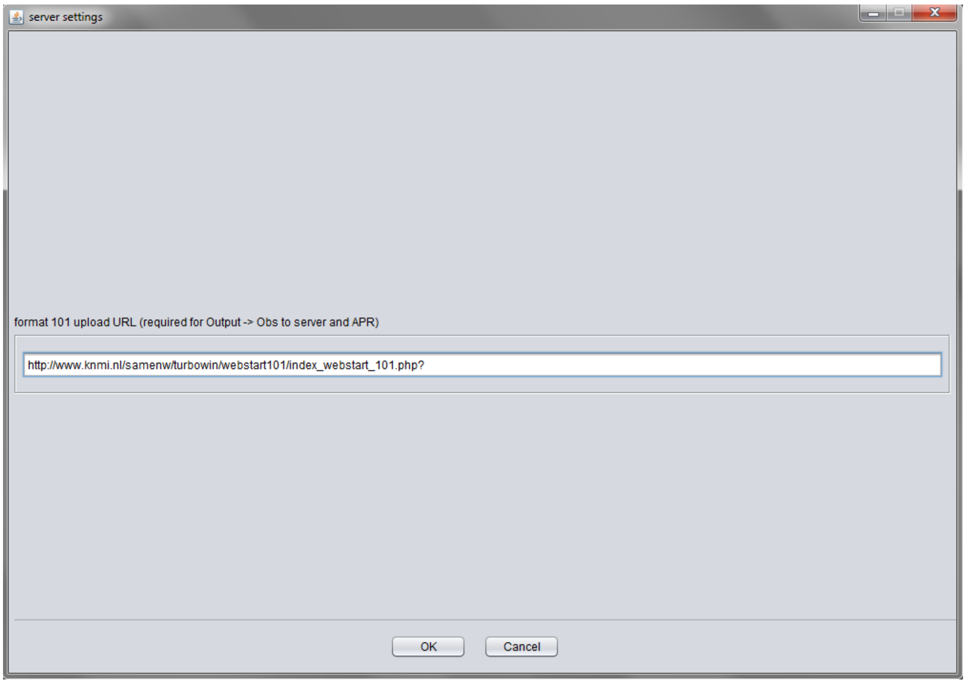


fig 56 TurboWin+ input screen to set upload URL (inserted URL as an example)

TurboWin+ will add the weather observation as key/value pair indicating the weather data. Key: "obs" the weather data itself is according format#101 and also encoded by TurboWin+. TurboWin+

translates (encodes) the format#101 observation into application/x-www-form-urlencoded format using the UTF-8 encoding scheme.

Example of a full upload string:

http://www.nationalmeteoservice.com/observations/turbowin/obs.php?obs=+TESTNLYP%40C%7F%7C%60XZ%40F%7D%7CdW%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7F%7C%40

note: So on the server the received observation string must be decoded e.g. with php function urldecode (Decodes any %## encoding in the given string. Plus symbols ('+') are decoded to a space character)

Response and Errors format 101

All requests must return a HTTP response (status) code. A success is indicated by 200. Anything else is a failure.

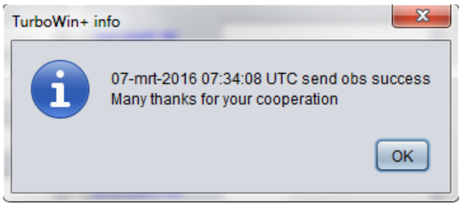


fig 58 Server-side HTTP response code 200 translated by TurboWin+ and shown as a pop-up message in TurboWin+

FM13 API

In TurboWin+ it is also possible to send a FM13 observation straight to a server of a National Meteorological Service or Centralized Service.

Information in this chapter 'Obs to server'

The information in this chapter is subject to change. For the latest info please send a mail to pmo.office@knmi.nl

KNMI scripts

On request the php scrips of the KNMI server-side implementation can be obtained from pmo.office@knmi.nl

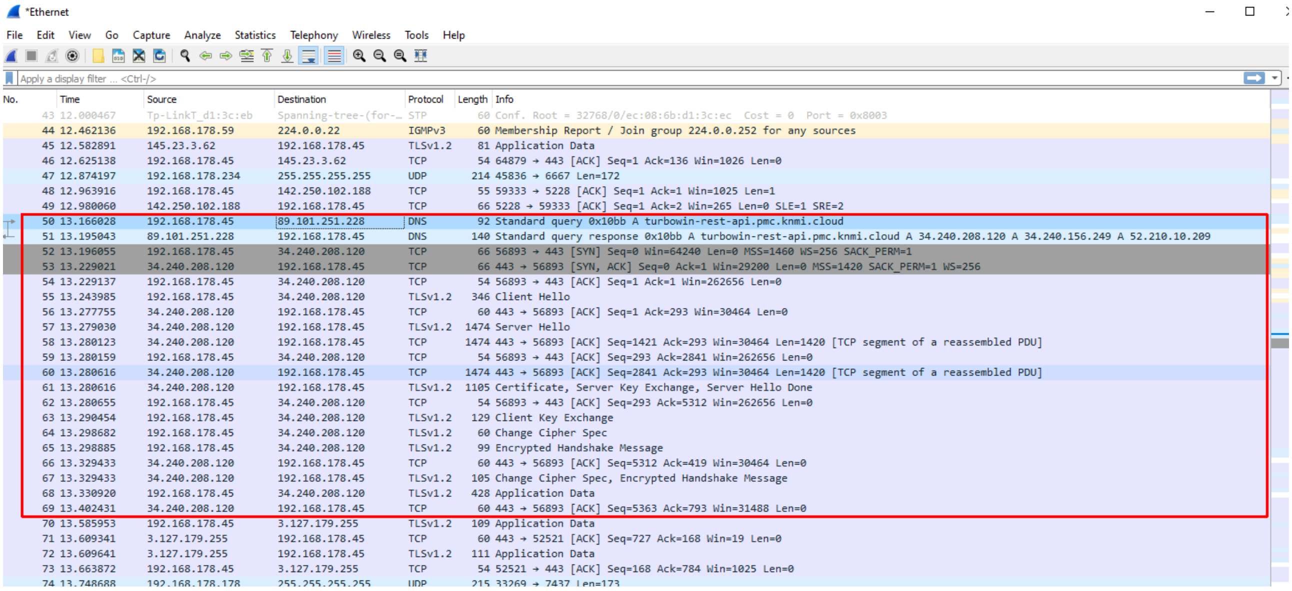
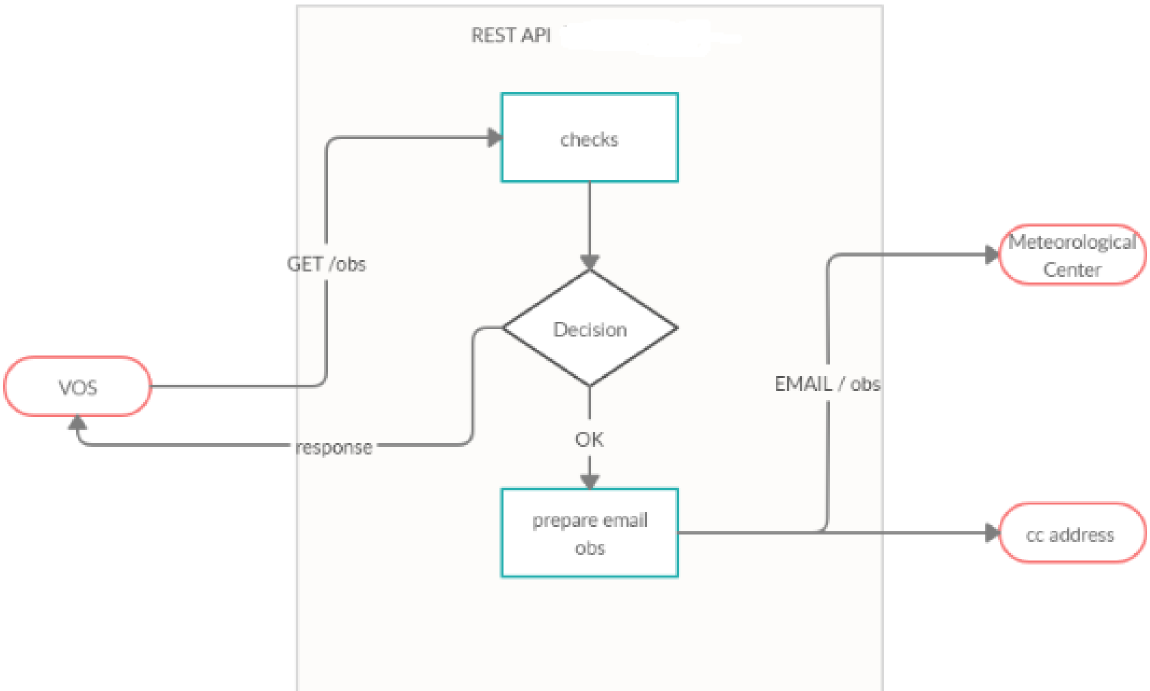


fig 59 example network monitoring trace TurboWin+ format #101 obs upload + response

Functional design



TurboWin+ rest api functional design

10.11 SRM (Serial communication Recovery Mode)

If an application has lost the serial communication with a device, most of the times you have to manually restart the system. Thanks to the in TurboWin+ integrated SRM (based on threading) most times it will recover automatically after loosing the serial connection. You could test this eg by removing the (serial)USB cable and after a while reconnecting the USB cable again. (Every hour [hh+43] minutes the serial connection will be tested and re-established if necessary)

```
04-Nov-2019 08:40:06 UTC [GPS] data listener thread cancelling
04-Nov-2019 08:43:06 UTC [GPS] starting-com-port closed (User-Specified Port)
04-Nov-2019 08:43:11 UTC [GPS] found serial com port (ok): Prolific USB-to-Serial Comm Port (COM12)
04-Nov-2019 08:43:15 UTC [GPS] found GPS on port: Prolific USB-to-Serial Comm Port (COM12)
04-Nov-2019 08:43:16 UTC [GPS] data listener thread restarting
04-Nov-2019 08:44:06 UTC [GPS] start receiving data again
```

fig 60 SRM logged

10.12 Check your observations

If internet available, you can check your sent observations on the E-SURFMAR VOS Control Panel (until a few days ago)

TurboWin+ select: Info → Statistics (internet).

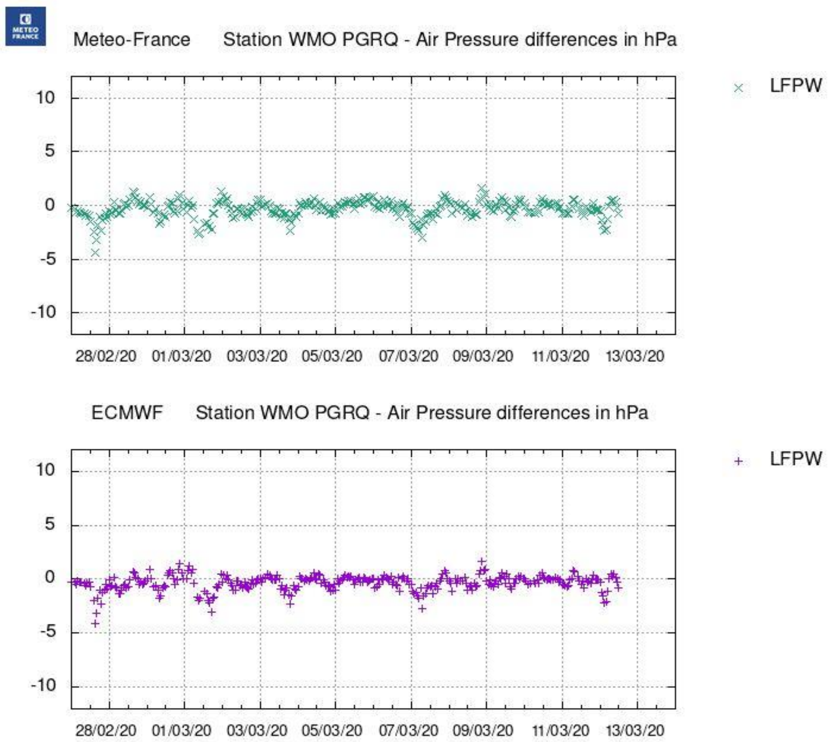


fig 61 example of air pressure quality control on the E-Surfmar VOS Control Panel

10.13a Autostart after reboot

After TurboWin+ installation you can instruct your OS to autostart the application after a OS reboot

On a computer with Windows OS:

1) Press the Windows key + R to open the Run box. Type **shell:startup** and press Enter. (It will reach the start-up folder)

2) Copy the TurboWin+ desktop icon (right click) and paste into the start-up folder. (or Right click in the Start-up folder for pop-up menu: New → Shortcut → Browse… → Select TurboWin+ → OK → Next → Finish)

3) Restart the computer to test it

4) Note, with the W10 settings menu you also can verify TurboWin+ autostart [fig 62 and fig 63]

10.13b Autostart after reboot and after sign in

If officers must log into their own account/virtual machine when they come on watch it is also possible to automatically start-up TurboWin+ after every log in.

On a computer with Windows OS:

1] "Turn On Automatically Restart Apps after Sign In": via the main (administrator) account: Settings -> Accounts -> sign-in- options -> Restart apps: ON (Available from at least Windows 10 build 18963)

2] for every account:

a) Press the Windows key + R to open the Run box. Type **shell:startup** and press Enter. (It will reach the start-up folder)

b) Copy the TurboWin+ desktop icon (right click) and paste into the start-up folder. (or Right click in the Start-up folder for pop-up menu: New → Shortcut → Browse… → Select TurboWin+ → OK → Next → Finish)

3) Sign out with the present account and sign in with a second account to test it

4) Note, with the W10 settings menu you also can verify TurboWin+ autostart [fig 62 and fig 63]

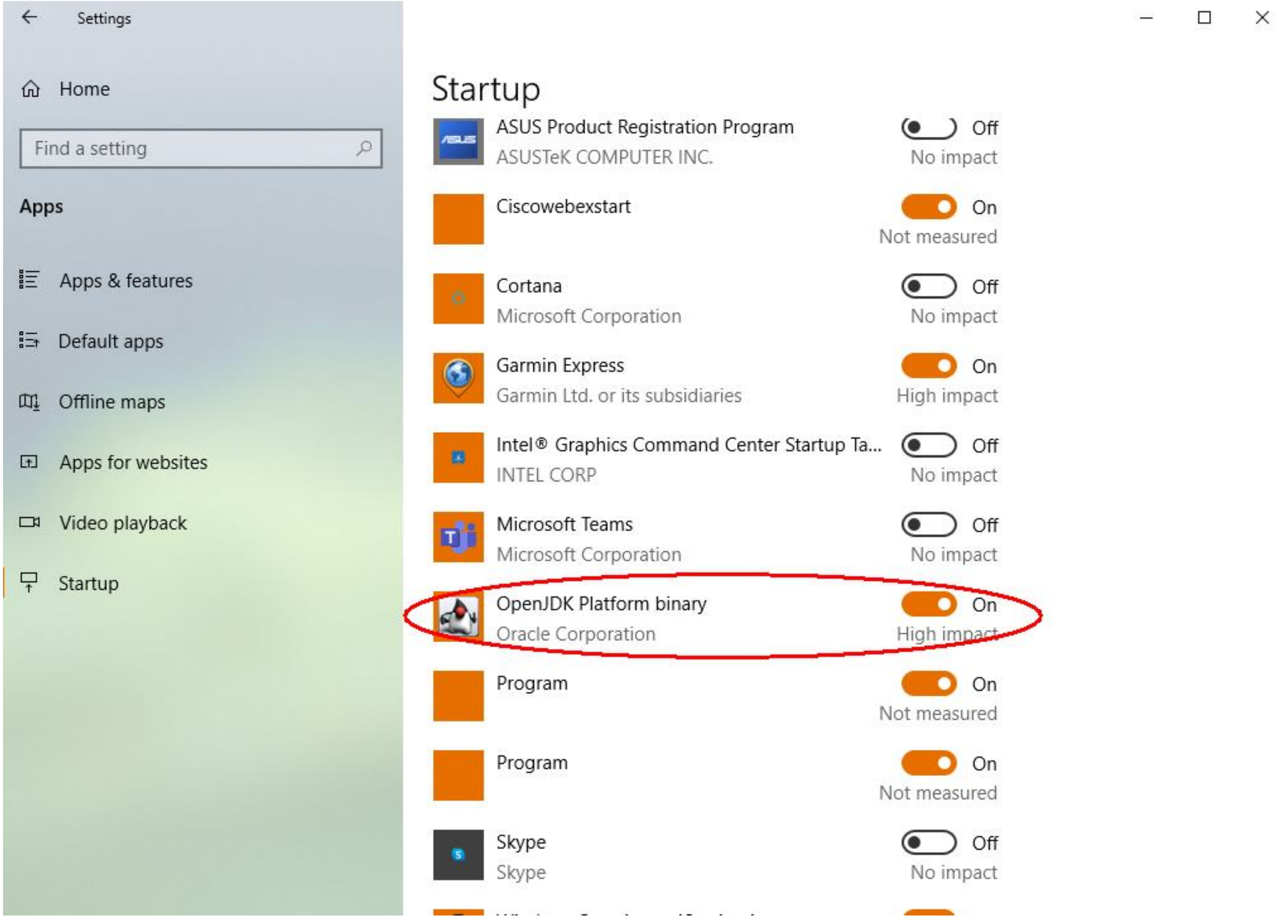


fig 62 Windows 10 64bit startup setting for TurboWin+ autostart

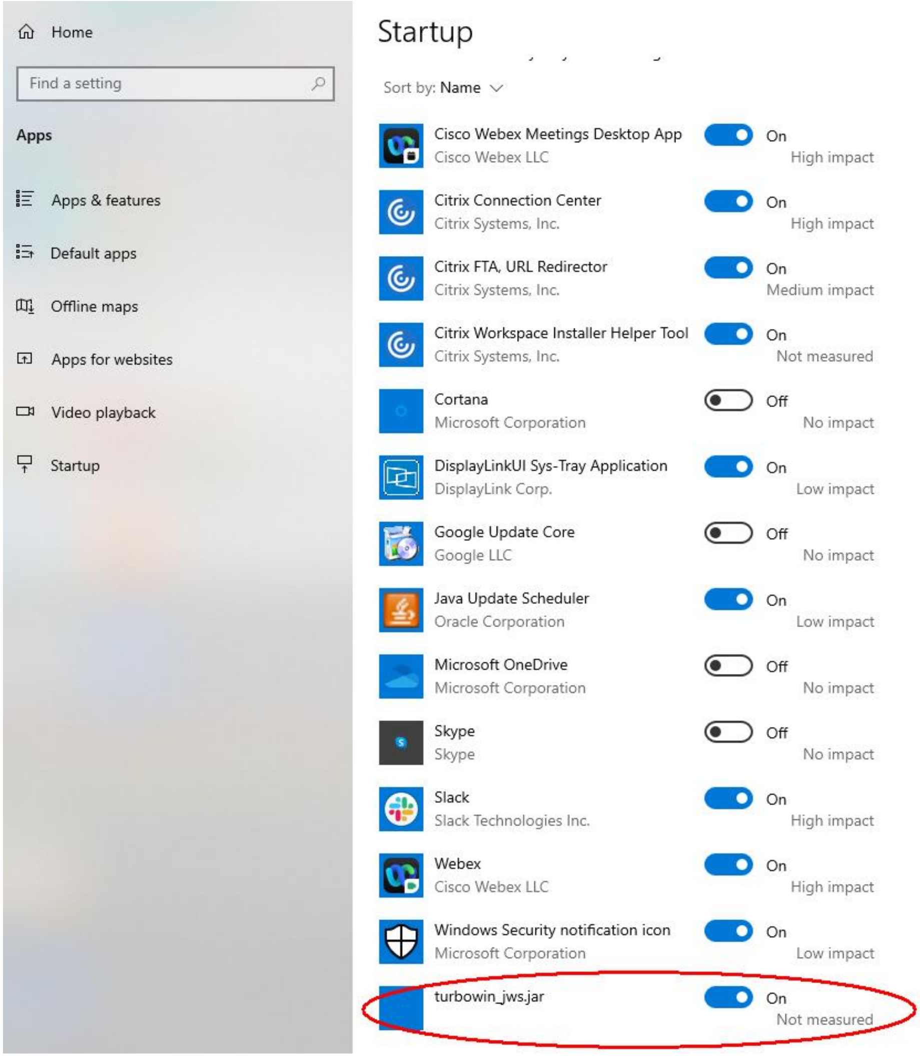


fig 63 Windows 10 32bit startup setting for TurboWin+ autostart

10.14] Test email Python module outside TurboWin+

For testing one can start the TurboWin+ (python) secondary email module outside TurboWin+.
e.g. with Windows PowerShell (see Fig 64 for an example) and check the results.
The email module is correctly configured by the system if the results are comparable as depicted in the example figure.

If not ok the console will show lines as “LoadLibrary: the specified module could not be found”
Cleaning the temp* folder might be sufficient to solve this problem.

**When started Python creates a temporary folder in the appropriate temp-folder location for this OS. The folder is named _MEIxxxxxx, where xxxxxx is a random number. The _MEIxxxxxx folder is not removed if the program crashes or is killed (kill -9 on Unix, killed by the Task Manager on Windows, “Force Quit” on Mac OS). Location temp folder e.g., C:\users\name\AppData\Local\Temp_MEIxxxxxx*

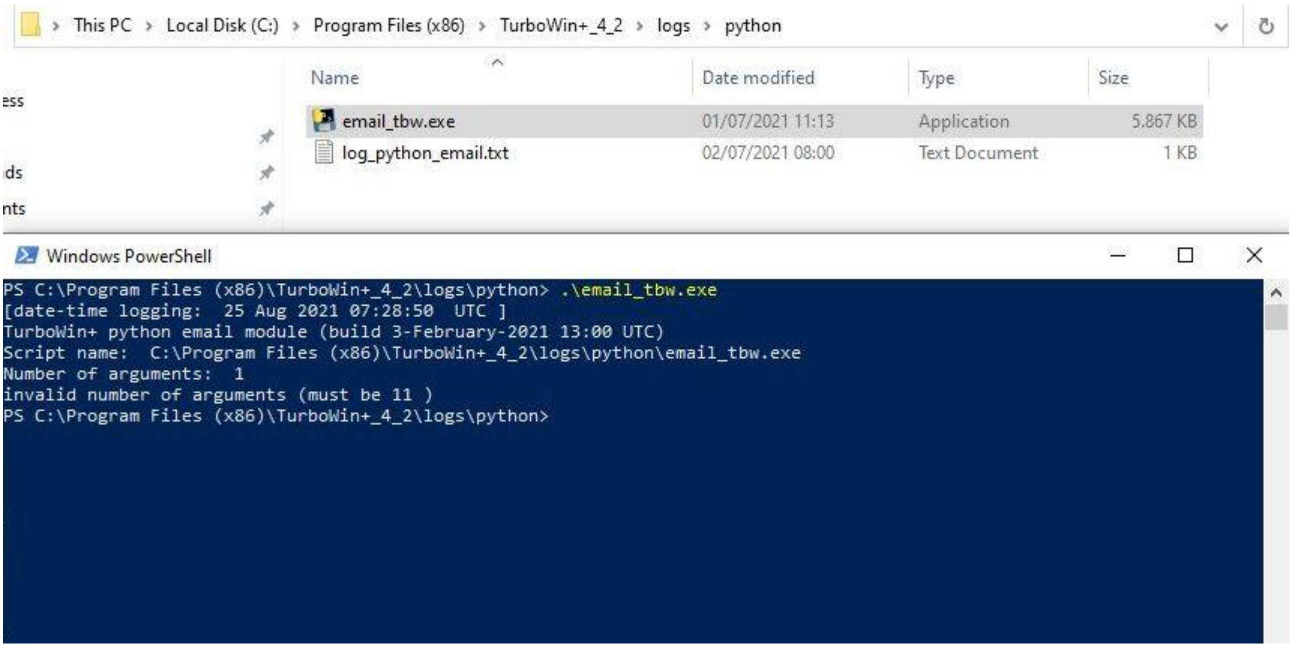


Fig 64 testing email module

10.15] Test “Obs to server” outside TurboWin+

The server connection can simply be checked and tested, even without running TurboWin+, by pasting the appropriate “Obs to server” URL in the address field of a random web browser. Result will be ‘int(700)’ if ok. If there are problems with the server (or onboard firewall blocking, no internet available etc.) the result will be something like ‘404 not found’

last updated: 06-February-2023